

Online Appendix

Trouble Downstream: How Rivers Carry Rainfall Shocks to Conflict in India

Mahiro Kobayashi[†]

September 29, 2026

[Click here for the latest online appendix](#) · [Main paper](#)

Contents

A Construction and Methods	2
A.1 Data and Sample	2
A.2 Choice of HydroBASINS Level	3
A.3 Measuring the Upstream Shock	3
A.4 The Exclusion Radius	7
A.5 Monthly Re-Indexing and Specification	9
A.6 Inference and Randomization: Design	11
A.7 Routed Water: Construction and Validation	15
A.8 Measuring What Arrives	15
A.9 Extent and Depth	15
A.10 Modeled Flooding versus the Flood Catalogue	16
A.11 Construction and Validation	16
B Additional Figures and Tables	18
B.1 Descriptive Figures	18
B.2 Main-Result Robustness	22
B.3 Mechanism	32
B.4 The Dam Arm	49
B.5 Public Employment	50
B.6 The Flood Arm	51

^{*}Graduate School of Public Policy, University of Tokyo.

[†]Last modified on September 29, 2026.

Appendix A. Construction and Methods

This appendix documents how the data, the treatment, and the routed-water measures are built, and the design of the timing and inference procedures. All result exhibits are collected in Appendix B.

A.1. Data and Sample

Boundaries and outcome.—All series are harmonized to 2001-census district boundaries (594 districts; the Jammu & Kashmir “data not available” catch-all unit is dropped). UCDP GED point events are assigned to districts by coordinates; the outcome is the calendar-year count of state-based, non-state, and one-sided violent events. Poisson fixed-effects estimation retains the 230 districts with a defined upstream catchment and at least one located event over 1990–2025 (8,280 district-years); incidence regressions retain all districts.

Treatment.—For each district, upstream level-10 sub-basins are found by tracing the HydroBASINS NEXT_DOWN pointer network to every basin whose flow path reaches the district. The excl.-neighbor rule then removes any basin intersecting the district or a bordering district. Basin monsoon rainfall is CHIRPS aggregated within basin polygons; the district treatment is the inverse-distance-weighted mean of Section III. A replication gate verifies the rebuilt exposure matches the panel column to numerical precision.

Sample window.—Table 1 of the main text reports the estimation sample on the longest window the data allow — 1990–2025. The binding constraint is the outcome rather than the treatment: the weather panel begins in 1981, UCDP GED in 1989, and the lag convention consumes the window’s first year.

The outcome is built from the UCDP Georeferenced Event Dataset, 1989–2025, 18,578 events in India, of which 16,626 are located to a district or better. Figure B.1 plots the record over time and Figure B.2 its composition.

Three features bear on the design. The series is dominated by a single episode: events peak at 1,550 in 2000, and Kashmir alone is 51.4 percent of all events. The geocoding improves sharply after the 1990s — 74.1 percent of events in 1989–1999 are located to a district or better, against 92.7 percent from 2000 — one reason the estimation window begins in 2000. And the precision restriction is close to neutral on composition: no conflict system or violence type shifts by more than 1.5 percentage points between all events and the located subset (the ticks in Figure B.2). Figure B.3 gives the cross-tabulation behind that statement — each system’s located events divided among UCDP’s three types of violence — which is what the decompositions of Tables B.3 and 3 condition on.

A.2. Choice of HydroBASINS Level

The HydroBASINS level must sit between two bounds: sub-basins small relative to districts, so upstream areas are delineated finer than the treatment unit, but nothing finer than the 0.05-degree CHIRPS grid. Level 10 sits inside that bracket with room on both sides — the median sub-basin covers 139 km² against a median district footprint of 4,060 km² (roughly thirty sub-basins per district) and is 4.9 times a CHIRPS cell, with only 7.2 percent of sub-basins finer than a single rainfall cell. It is also where the decomposition saturates: level 9 to 10 nearly doubles the units (22,471 to 41,915), while level 11 adds under a tenth as many again and leaves the median essentially unchanged. Level 10 is the last level that buys resolution, not merely the last that is coarse enough.

Figure A.1 shows what that granularity means on the ground: 21,622 level-10 sub-basins over India as a whole, and, in close-up, the Kosi–Ganga confluence, where a single district’s upstream catchment is assembled from dozens of sub-basins. The construction never operates at the district’s own scale — it aggregates these units along the drainage network.

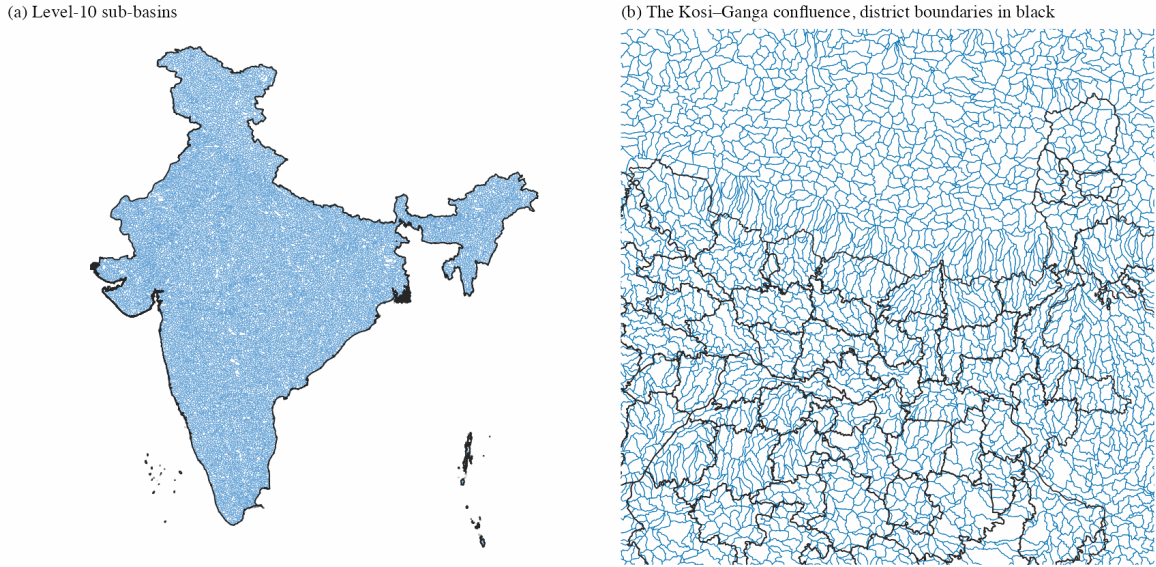


Figure A.1: The HydroBASINS level-10 delineation

Notes: Panel (a): level-10 sub-basin boundaries (blue) within India’s bounding box, national outline in black; boundaries are simplified for rendering at this scale. Panel (b): the Kosi–Ganga confluence at full resolution, with 2001-census district boundaries in black. Sub-basins are the units on which upstream rainfall is accumulated before it is assigned to districts through the drainage network.

A.3. Measuring the Upstream Shock

Which rainfall product.—The headline is estimated on CHIRPS v3, and Figure A.2 re-estimates it changing nothing but the precipitation series, under the paper’s own specification — the full control set including upstream temperature and the spatial

block — on identical rows (3,382 district-years on 178 districts; the window is 2000–2018 because the alternative aggregates exist only there, so the CHIRPS v3 point is the headline *specification* on that sample, not the paper’s headline number). The elasticity is 1.98 on CHIRPS v3 and positive on every alternative, but only CHIRPS v2 approaches it: 1.09 ($p = 0.062$), against 0.39 on GPCC, 0.22 on CRU, 0.30 on ERA5-Land ($p = 0.421$) and 0.12 on TerraClimate. The sharpest comparison is internal — CHIRPS v2, the same product one version earlier, correlates only 0.69 with v3 after fixed effects and carries roughly half the coefficient. Either v3 resolves upstream monsoon rainfall that coarser or model-based products miss — what its station blending and resolution are for, in gauge-sparse transboundary catchments — or the v2-to-v3 difference is noise that happens to correlate with conflict; this paper cannot separate the two. The magnitude is therefore a CHIRPS v3 magnitude, and readers who weight the other products should read the headline as a CHIRPS v3 result rather than a rainfall result.

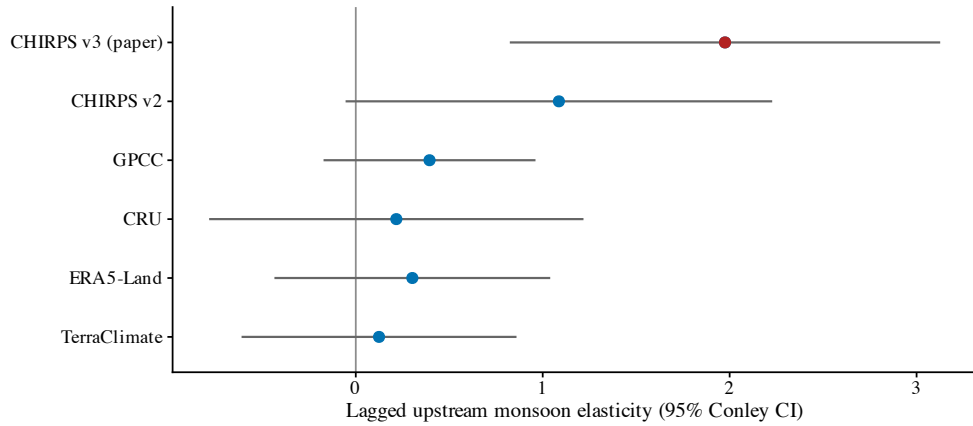


Figure A.2: The lagged upstream elasticity across precipitation products

Notes: Each point re-estimates the headline specification — Poisson, district and year fixed effects, upstream temperature, local weather and the spatial block, current and lagged — replacing only the precipitation series used to build the upstream and local rainfall terms. All rows share one sample: the 3,382 district-years (178 districts, 2000–2018) on which every product is observed. Bars are 95 percent intervals on Conley spatial-HAC standard errors (150 km). The CHIRPS v3 point (red) is the paper’s specification on this restricted sample rather than the headline estimate itself.

Where the treatment is measured.—23 percent of the upstream basins lie outside India, mostly in China, Nepal and Bhutan, where CHIRPS’s station coverage is sparsest. Rebuilding the exposure from Indian basins only — renormalising the inverse-distance weights, not zeroing out terms — gives 0.67 (standard error 0.44) on 164 districts, smaller than the headline and no longer significant. Excluding basins above 2,500 m gives 0.75; excluding everything above 1,500 m gives 1.26 (standard error 0.36), *larger* and better determined than the headline. The non-monotonicity is the honest summary: the result does not live entirely in the badly-measured high catchment, but it is sensitive to which foreign basins enter the treatment — and dropping basins changes

the estimand as well as the measurement, since snowmelt-fed basins deliver water on a different schedule from plains basins.

Where the gauges are.—The blending is not uniform, and the product’s own diagnostics say where it happens: CHIRPS ships the per-pixel count of stations it blended each month, and Figure A.3 maps the monsoon-month average over the upstream basins. The geography is not the one elevation and nationality proxies suggest. 92 percent of the 21,733 upstream basins average fewer than one blended station per twenty pixel-months, and the *Indian* basins are the worst covered — 96 percent effectively ungauged against 82 percent of foreign basins — because the IMD network does not flow into the product while Nepal’s does. Over these catchments CHIRPS is close to a satellite product with a Nepali gauge anchor, and the median estimation district draws 97 percent of its upstream weight from pixels carrying no station at all. Panel B states this in the design’s own units: the gauge support of the treatment, the exposure weights applied to station density.

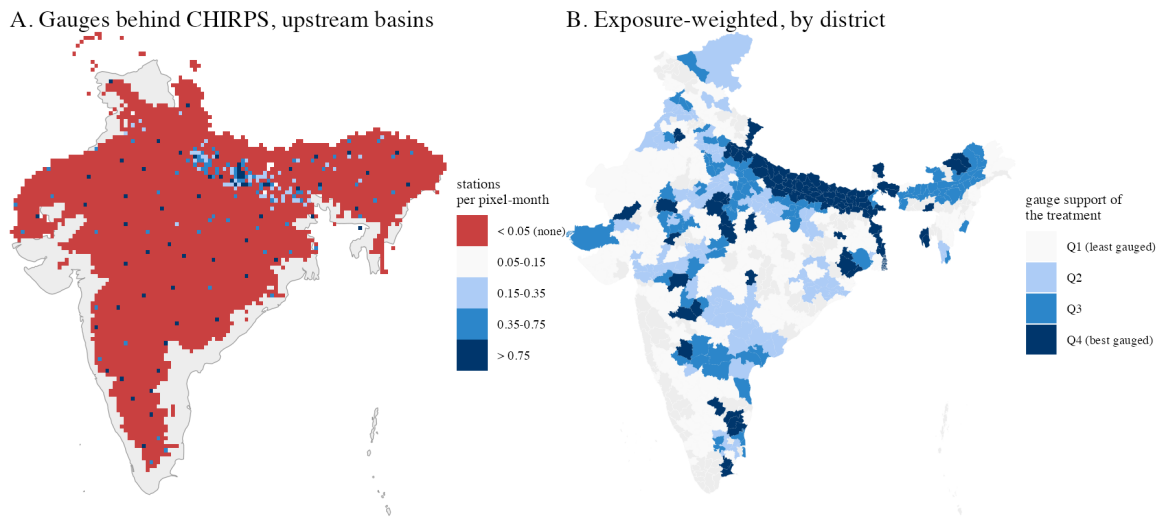


Figure A.3: The gauges behind the treatment

Notes: Panel A maps CHIRPS v3’s own station-density diagnostic — the number of gauges the product blended, per 0.25-degree pixel — averaged over monsoon months (June–September) 1990–2025, masked to the upstream basins of the excl.-neighbor crosswalk. Red marks pixels averaging fewer than one blended station per twenty pixel-months. Panel B aggregates the same density to districts with the inverse-distance weights that build the upstream treatment, so it reads as the gauge support of each district’s exposure; quantile bins, darker better gauged. Grey districts have no upstream catchment in the construction.

Against the gauge-based grids.—Figure A.4 compares the product against the two gauge networks it does not contain. Panel A compares district monsoon totals with the IMD 0.25-degree gauge-based grid (Pai et al., 2014) — the authoritative network over India, and one whose grid stops at the border, outside which 23 percent of the upstream basins lie. Across 589 districts the two products’ monsoon totals correlate 0.66 within district at the median, with a median absolute gap of 14 percent of the

IMD total. Panel B makes the comparison the treatment actually needs — upstream basins, including the foreign catchments IMD cannot see — against GPCC, the gauge-based product that is global: within-basin correlations are 0.70 (Indian basins) against 0.65 (foreign), with median absolute gaps of 11 and 21 percent. The disagreement concentrates along the Himalayan arc, where Figure A.3 shows the gauges to be missing — which is the measurement fact the restrictions above test and the terciles of Table A.1 price.

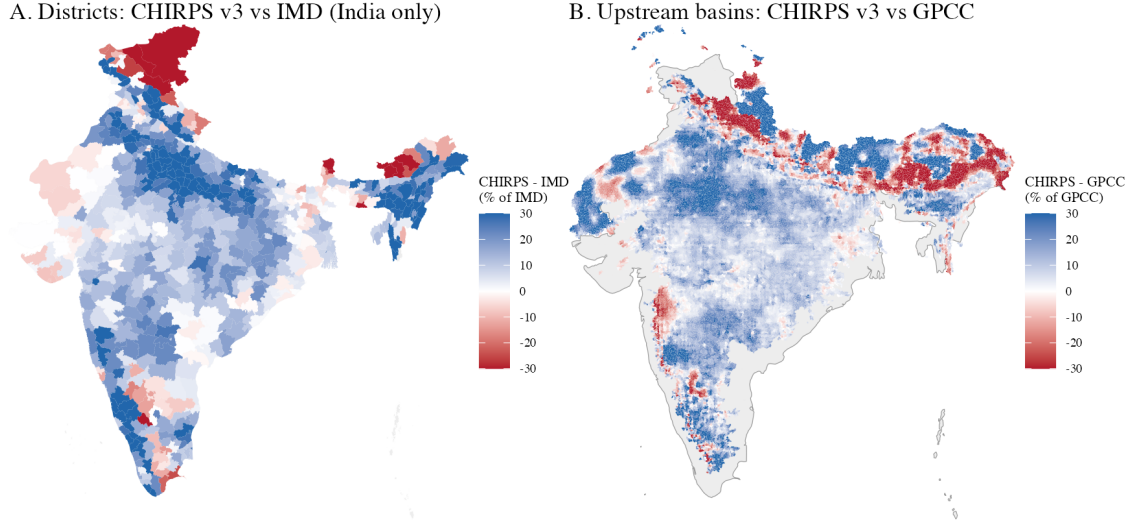


Figure A.4: CHIRPS v3 against the gauge-based grids

Notes: Panel A: the district’s mean June–September CHIRPS v3 total (the panel’s own local-rainfall measure) against the same total from the IMD 0.25-degree gauge-based grid, 1990–2025, as a percentage of the IMD total; blue means CHIRPS reads wetter. IMD covers India only, so the comparison cannot reach the foreign upstream basins. Panel B: the same comparison at basin level against GPCC v2022, the gauge-based product with global coverage, over the upstream basins of the treatment, 1990–2020. Gaps beyond ± 30 percent are censored to the scale’s ends.

Where the gauges are, tested.—Table A.1 asks whether the headline lives in badly-measured exposure, splitting districts into terciles of the gauge support of the treatment. It does not: the least-gauged tercile carries 1.454 against 1.928 in the best-gauged, both significant, and the formal interaction finds no detectable heterogeneity by gauge support ($p = 0.48$ for the least-against-best contrast). Classical measurement error would attenuate the estimate precisely where gauges are scarce, so a coefficient that does not fall there is evidence against the reading that the result is an artifact of poorly measured rainfall. The hard-exclusion arm — dropping the least-gauged tercile outright — keeps a positive but imprecise 1.014 (standard error 0.757) on 148 districts: like the India-only restriction above, exclusion answers the power question, not the measurement one, which is why the splits and the interaction carry the test.

Table A.1: The headline estimate by gauge support of the treatment

	Full sample	Drop least-gauged third	Least gauged (T1)	Middle (T2)	Best gauged (T3)
log Upstream Precip. ($t - 1$)	1.410*** (0.329)	1.014 (0.757)	1.454*** (0.516)	-0.698 (0.468)	1.928*** (0.731)
Observations	8,280	5,328	2,952	2,736	2,592
Number of Districts	230	148	82	76	72

Notes: Poisson fixed-effects on the specification of Table 2, column 4, 1990–2025. Districts are split into terciles of the gauge support of the treatment: CHIRPS v3’s own monthly station counts (Figure A.3), averaged over each district’s upstream basins with the inverse-distance weights that build the exposure. The formal test is the interaction, which lets the treatment slope differ by tercile while holding every other coefficient common: the least-gauged tercile differs from the best-gauged by 0.55 ($p = 0.48$) and the middle by -0.18 ($p = 0.84$) — no heterogeneity by gauge support is detectable. The tercile columns, in which every coefficient may differ, run on 72–82 districts each and are shown for transparency; their movement is what samples of that size produce. Conley spatial-HAC standard errors (150 km cutoff) in parentheses, which is the paper’s headline estimator and the basis of the significance stars; Online Appendix Figure B.7 re-prices the headline coefficient under alternative variance estimators. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

A.4. The Exclusion Radius

The exclusion radius.—Figure A.5 reports the sweep itself — the elasticity at every radius, under both specifications and both distance metrics, on the varying and the fixed district set. The design fails in opposite ways at the two ends of that continuum (Section IV), and both bounds are measurable on the same cells. Push the ring *in* and the treatment stops being rain from elsewhere: its post-fixed-effects correlation with the district’s own monsoon rainfall rises from 0.58 at 100 km to 0.87 with no exclusion, while the fixed-sample elasticity rises from 0.63 to 1.13, because at short radii the treatment is partly local weather. Push it *out* and the map stops carrying information: the ratio of the true estimate to the 95th percentile of within-state map permutations falls from 1.85 to 0.98 at 100 km, where the true map no longer clears its own placebo. These eleven radii are nested subsets of one catchment, not independent tests; the exact p -values are in Section A.6.

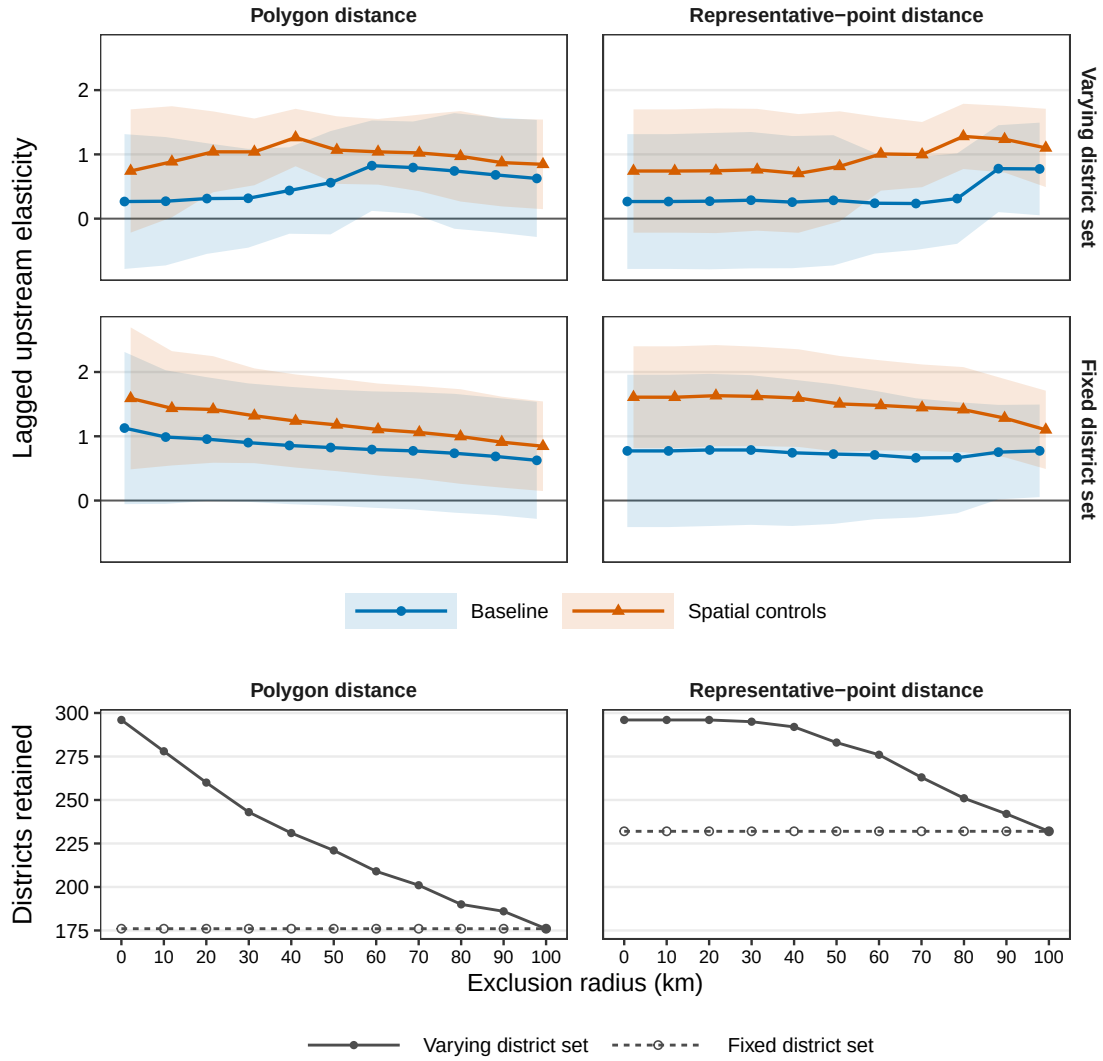


Figure A.5: The lagged upstream elasticity across a continuous exclusion radius

Notes: Each panel plots two specifications of Table 2 — column 1, district and year effects alone (Baseline), and column 4, which adds the spatial block (Spatial controls) — re-estimated at every radius, so the only thing varying along each line is where the catchment boundary is drawn. Columns of panels are the two distance metrics: polygon distance from the district to the basin, and distance to the basin’s representative point. Rows are the two sample arms: the varying arm lets the district set change as wider buffers strip catchments, while the fixed arm holds it at the districts surviving the 100-kilometre cutoff, so movement along it is not composition. Ribbons are Conley (150 km) 95 percent confidence intervals. Exposure and the upstream temperature control are rebuilt at every radius. The lower panels give the districts each cell retains, which is what the widening bands at large radii reflect. The radius is a buffer on how far a basin must lie from the district to count: at zero every basin in the district’s upstream catchment enters, and at 100 kilometres only those more than 100 kilometres away are kept, measured by the distance rule of the panel’s column. The paper’s own exclusion is not a distance rule — it removes basins that touch the district or any district bordering it — so no cell of this figure reproduces Table 2. The two bounds quoted in the text are computed on this figure’s own Baseline, polygon-distance, varying-sample cells — the same window, outcome, controls and districts, asserted cell for cell when the series are built — and carry no spatial block, because both are diagnostics about the treatment and conditioning on neighbours’ conflict and weather would change what the diagnostic measures. The first is the correlation, after district and year effects, between the lagged upstream treatment and the district’s own lagged local monsoon rainfall; the second compares the estimate under the true district–catchment map with the 95th percentile of estimates from 299 within-state permutations of that map at each radius. Neither bound is a reason to move the exclusion: the radius was fixed ex ante, and the sweep was run afterwards.

Who survives the buffer.—The districts retaining a catchment at 100 km are not a random subset: their upstream catchments are twenty times larger (62 against 3.1

thousand km²), their basins more distant (148 against 12 km), their river cells far flatter (0.91 against 0.56 flat share) — large-floodplain districts whose water genuinely comes from far away, and not disproportionately Northeastern. The composition effect is the paper’s own thesis restated as heterogeneity.

A replication note.—The upstream temperature control must be rebuilt at each radius alongside the treatment: holding it fixed while the treatment moves silently changes the specification, shifting the 50 km estimate by 0.06 without any error being raised.

A.5. Monthly Re-Indexing and Specification

The main text dates the treatment to a year; this subsection asks when within the following two years the response appears, and whether it is specific to the monsoon or merely to rainfall.

Re-indexing.—A district-month is relabelled by monsoon year v (June to May) and position $k \in \{0, \dots, 11\}$, months since June — a complete relabelling of the calendar, with $y = v + \mathbb{1}\{k \geq 7\}$. Let C_{ivk} count events UCDP places in district i in that month, located to a district and dated to a week or better.

Specification.—Let $R_{i,v}^M$ be upstream monsoon rainfall — exactly the main text’s treatment — and $R_{i,v}^N$ the complementary January–May and October–December total, both one number per district-year. With T^M and T^N the matching upstream temperatures,

$$\begin{aligned} \log \mathbb{E}[C_{ivk}] = & \beta_k^M \log R_{i,v}^M + \beta_{k+12}^M \log R_{i,v-1}^M + \beta_k^N \log R_{i,v}^N + \beta_{k+12}^N \log R_{i,v-1}^N \\ & + \gamma_k^M T_{i,v}^M + \gamma_{k+12}^M T_{i,v-1}^M + \gamma_k^N T_{i,v}^N + \gamma_{k+12}^N T_{i,v-1}^N \\ & + \mathbf{x}_{ivk}' \boldsymbol{\theta} + (W \mathbf{z}_{ivk})' \boldsymbol{\psi} + \alpha_i + \lambda_{vk}, \end{aligned} \quad (1)$$

estimated by Poisson pseudo-maximum-likelihood with standard errors clustered on district. Each coefficient vector has twenty-four elements: an observation at position k uses element k , the season that opened its own monsoon year, and element $k + 12$, the season before. This is one regression, not twenty-four — estimating it month by month would lose more than half the observations, since a Poisson drops districts with no conflict in the month being estimated.

What varies, and what does not.—The rainfall on the right-hand side is a seasonal total, so what changes across the twenty-four coefficients is which month’s violence sits on the left; month-to-month movement in the estimates is not structure. The twenty-four positions all fall after the monsoon begins, so the design contains no pre-shock months and offers no placebo lead.

Controls and fixed effects.— $\mathbf{x}_{ivk}$ holds the district’s own rainfall and temperature

that month and the month before; $\mathbf{W}\mathbf{z}_{ivk}$ holds the queen-contiguity averages of neighbours' previous-month conflict and of neighbours' own weather, rainfall in logs on both sides. The contemporaneous spatial lag is omitted for the reason Table B.6 gives, and the Poisson carries no own lagged conflict term. The time effect is $\lambda_{vk} = \delta_{vk}$, one parameter for every calendar month in the sample (425 of them): a single month-year effect absorbs anything common to India in that month — an election, a national operation, a change in reporting — where separate month-of-year and year effects absorb only the average season and the average year. It does not absorb the horizon structure, because within a month-year cell every district sits at the same two horizons, so β_k and β_{k+12} are identified from variation across districts in the two vintages' rainfall. This is the strictest time control the design admits. Two rungs are reported, without the spatial block and with it.

Result.—Figure A.6 plots both rungs over 1990–2025, the window of Tables 2–B.2. Monsoon rainfall is flat or negative through the season itself and runs positive through the following year: summed over months seven to eighteen it gives 4.98 under month-year effects alone and 1.66 with the spatial block, the peak falling in November of the following year (0.65). Non-monsoon rainfall, which is nearly orthogonal to monsoon rainfall once the fixed effects are removed, does nothing over the same window in either rung: -3.07 and -1.29. The delayed response is a monsoon phenomenon rather than a rainfall phenomenon, which is what the agricultural mechanism requires.

What the spatial block carries.—The two rungs differ in size rather than shape. Adding neighbours' lagged conflict and neighbours' weather cuts the monsoon sum by roughly two thirds while leaving the profile — negative in season, positive from the turn of the year, peaking in the autumn of $t + 1$ — intact. Part of what the monthly profile picks up without the block is therefore neighbourhood diffusion rather than the district's own response, and the spatial rung is the one the tables' column 4 corresponds to and the one the text quotes.

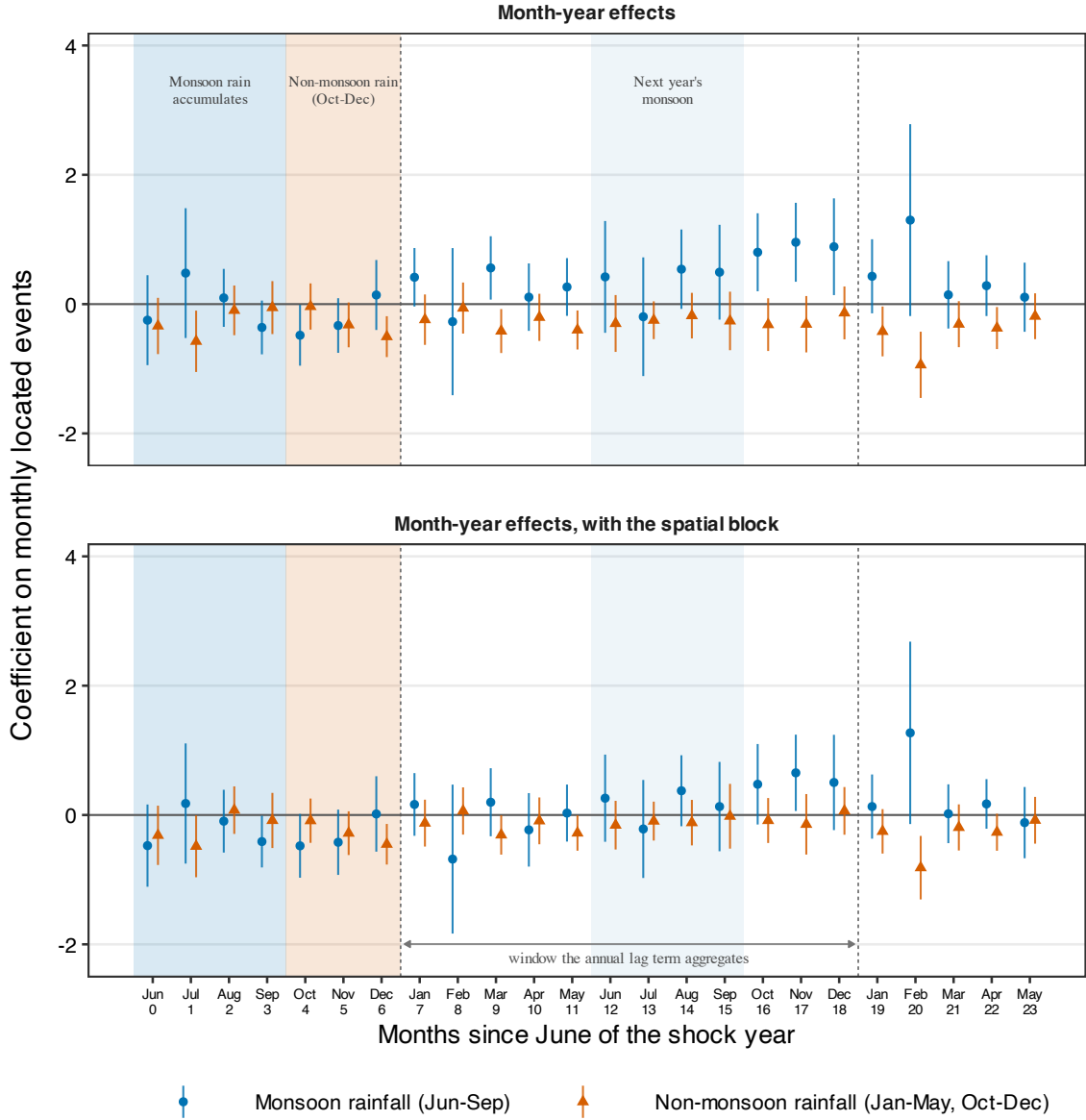


Figure A.6: Monthly horizons of the upstream-rainfall shock, by season

Notes: Each panel plots $\hat{\beta}_h^M$ and $\hat{\beta}_h^N$ from one estimate of equation (1), both seasons carried simultaneously at all twenty-four horizons, against months since June of the shock year. Points are Poisson coefficients with 95 percent cluster-by-district intervals. The panels differ only in the spatial block; both carry district and month-year effects, and 425 month-year cells span 1990–2025. Shaded bands mark the months in which each rainfall measure is accumulated. The monsoon measure is the June–September total of the shock year, so coefficients inside the first band are contemporaneous with the rain rather than responses to it. The second band is the October–December portion of the non-monsoon measure; its remaining five months, January–May of the same calendar year, fall before horizon 0 and lie off the left edge, so most of the non-monsoon treatment is accumulated before the monsoon it is raced against. The third band is the following year’s monsoon, which falls inside the window the annual lag term aggregates (horizons 7–18, between the dashed rules): a lagged annual coefficient cannot separate its own shock’s late response from the next monsoon’s early one, and the monthly profile is what makes that visible. Estimated on 94,350 district-months over 222 districts; the rung without the spatial block runs on 94,350 over 222. Identity is carried by shape as well as colour.

A.6. Inference and Randomization: Design

Districts along the same river share upstream basins, so their treatments are mechanically correlated and district-clustered standard errors are too small (Adão et al.,

2019).

How many shocks are there?.—Shift-share consistency is asymptotic in the number of shocks. The effective number, $1/\sum_b \bar{s}_b^2$, is 1,339 across level-10 sub-basins, 39.5 across major sub-catchments, and 7.64 across macro basins; the coarse count is the one inference should use, which is why the ladder below runs to the macro basin. Identification is not concentrated in a few places either: 108 effective districts of 176, the ten largest contributing 18.4 percent, no conflict theatre exceeding 45 percent.

Recentering on expected exposure.—Recentering the treatment on expected exposure μ_i is the [Borusyak et al. \(2022\)](#) correction. When climatology is time-invariant, district effects absorb μ_i exactly, and recentering reproduces the baseline 1.410 to three decimals (1.410) — a check on the construction rather than a result. The demanding version lets μ_i differ by decade; expected exposure then carries 2.9 percent of exposure variance and the estimate attenuates to 1.280 (standard error 0.389, $p = 0.001$) — a ninth of its size, still significant at one percent. Table [A.2](#) collects these diagnostics with the shock counts and concentration measures.

Figure [B.7](#) re-prices the *same* regression — the specification of Table [2](#), column 4, on 1990–2025 — under progressively coarser dependence structures, plotted as test statistics so every estimator reads directly against the conventional thresholds. The estimate does not depend on fine-grained spatial independence: it clears one percent under every estimator, including clustering on 22 macro-basins covering entire river systems and a spatial-HAC kernel at 500 kilometres, with the standard error rising from 0.275 under district clustering to at most 0.389 — a little under half as large again, the price of allowing dependence across whole river systems. The least favourable cell in the whole exercise is Conley at 450 kilometres, $t = 3.62$, against a one percent critical value of 2.576. The Conley curve is not monotone in the cutoff, and at some cutoffs the variance matrix must be repaired before use; the level of the curve is informative and its wiggles are not.

Clustering on shared catchments.—The grouping the design actually implies can be built from the treatment itself: cosine similarity between districts’ upstream weight vectors, thresholded at 0.50, gives 90 connected components linked by shared upstream water (Figure [A.7](#)). Re-clustering the headline on those groups inflates the standard error from 0.275 to 0.319 — 1.16 times, within a thousandth of state clustering — and the coefficient, 1.410, remains significant far below one percent. Clustering on a partition estimated from the same exposure that enters the regression is not free, so this rung is a bound rather than a replacement; Figure [B.7](#) places it alongside the others on the same regression.

Table A.2: Shift-share design diagnostics

<i>A. How many independent shocks?</i>		
	Units	Effective number
HydroBASINS level-10 sub-basins	18,413	1,340
PFAF level-5 sub-catchments	93	39.5
PFAF level-4 macro basins	24	7.64
largest single macro-basin share	—	0.31
<i>B. Is identification concentrated?</i>		
	Effective districts	Top-ten share (%)
Time-invariant σ_i^2	140	15.9
σ_i^2 by decade	150	14.6
<i>C. Where does that variation sit?</i>		
	Districts	Share of σ^2 (%)
Other	64	40.4
Maoist corridor	94	29.0
Kashmir	34	21.6
Northeast	38	9.0
<i>D. Recentering on expected exposure</i>		
	Coefficient	Std. error
Baseline	1.410***	(0.275)
Recentered, time-invariant μ_i	1.410***	(0.275)
Recentered, μ_i by decade	1.280***	(0.389)
μ_i share of exposure variance	2.9%	—

Notes: All panels are computed on the estimation sample of Table 2, column 4 — 8,280 district-years over 230 districts, 1990–2025. Panel A gives $1/\sum_b \bar{s}_b^2$ over average exposure shares: the number of *independent* shocks the design has at each level of the basin hierarchy, which is what shift-share asymptotics run in. The sub-basin count should not be believed — sub-basins inside one monsoon system are not independent draws — which is why inference is clustered at the macro basin. Panel B reports the inverse Herfindahl of σ_i^2 , the within-district variance of the treatment: the effective number of districts contributing identifying variation, and the share carried by the ten largest. Panel C splits that variation by conflict system, districts assigned to the system carrying most of their located events and collapsed to the four categories used previously; ‘Other’ is a residual, not a theatre. Panel D recenters the treatment on expected exposure μ_i . With μ_i time-invariant, district effects absorb it exactly and the estimate cannot move — reproducing the baseline is a check on the construction, not a result. Letting μ_i differ by decade is the demanding version. Standard errors clustered by district. ** 0.05, *** 0.01.

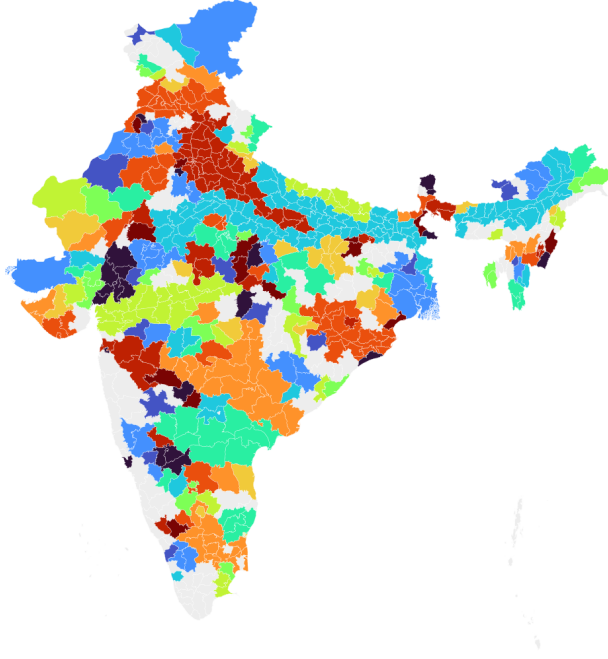


Figure A.7: Districts grouped by shared upstream catchment

Notes: Connected components of the district-by-district cosine similarity in upstream exposure, thresholded at 0.50. Exposure is the inverse-distance weight vector over HydroBASINS level-10 units that builds the treatment, so two districts are joined when more than 0.50 of their upstream water is shared, directly or through a chain of such links. Colours distinguish adjacent groups and carry no order. Districts drawn in flat grey have no upstream catchment in the construction. The partition is used in Section A.6 as a clustering level for standard errors, not as an estimation sample.

Randomization inference.—

The most demanding check reassigns the treatment map. Within each state, the mapping from districts to upstream-catchment rainfall series is permuted — each district receives the full time series of another district’s catchment — and the complete specification of Table 2, column 4 is re-estimated, 5,000 times. Figure B.8 shows the result, and panel (a) of Figure B.9 repeats it beside the timing tests. Not one permutation reaches the actual estimate of 1.41 (exact $p = 0.0002$); the placebo distribution is centred on -0.050 with a 95th percentile of 0.34.

Permuting within states makes this a demanding test: a counterfeit map still hands each district rainfall correlated with its own, so the question is not whether upstream rainfall matters but whether *this* district’s catchment matters more than a plausible neighbour’s.

Timing is the orthogonal question: the map permutation shuffles space and holds time, this one shuffles time and holds space. One permutation of the year labels is applied identically to every district, so each year’s cross-section moves as an intact block — districts sharing basins keep their correlated shocks — and only the shock–outcome alignment breaks. Panels (b) to (d) of Figure B.9 report it: the observed statistic sits outside the null on every version, $p = 0.0048$ on the coefficient and

$p = 0.0004$ studentized, against an actual t of 5.13. Permuting only *within* decades, which preserves trend and relaxes the exchangeability the free permutation assumes, leaves the answer unchanged ($p = 0.0004$); the score statistic implied by the Poisson likelihood agrees ($p = 0.0082$ free, $p = 0.0010$ within decades).

A.7. Routed Water: Construction and Validation

A.8. Measuring What Arrives

The treatment is rainfall hundreds of kilometres away; the district experiences whatever the river delivers. To observe that delivery directly I route two decades of ERA5-Land runoff through CaMa-Flood (Yamazaki et al., 2011), a global river routing model run at 6-arcminute resolution over South Asia for 2000–2018, in two configurations: a baseline carrying all water, and a masked ensemble in which each district’s own and surrounding runoff input is zeroed, so the routed series at the district’s main river cell carries *upstream-origin water only*. From the masked series I count *flood-stage days*: days on which the 3-day-mean discharge exceeds the cell’s own 2-year return level during the monsoon. Upstream rainfall generates these days mechanically — a log-point of upstream monsoon rain adds 2.3 flood-stage days (s.e. 0.24). Section VI reports the race between the rainfall total and these routed days (Table 4).

A.9. Extent and Depth

Inundated *area* and flood *depth* — the margins an optical flood map measures — carry no effect and absorb none of the flood-stage coefficient. This is not an artifact of measurement scale: rebuilt like-for-like, as days on which the district’s flooded fraction exceeded its *own* return level, the footprint remains a tight zero, alone and raced against discharge on identical observations. Depth so normalized does move conflict, but depth-days and discharge-days correlate at 0.94 within district — two readings of the same river — so it corroborates rather than competes. Nor is weighting the explanation: rebuilt weighting each pixel by population or kharif harvested area, the footprint still carries nothing.

The footprint measures are also validated on their own terms: on the Dartmouth catalogue’s impact fields across 144 usable India events, a log point of flooded area raises recorded deaths by 0.32 and displacement by 0.73. The measures that best predict who dies predict nothing about conflict, while routed discharge days continue to — humanitarian severity and conflict relevance are different quantities.

The footprint’s null is adaptation, not irrelevance: under abnormal inundation land moves *into* flood-tolerant rice and *out* of waterlogging-sensitive cash crops — the flood-tolerance margin of Emerick et al. (2016). Spread water is absorbed by land use; the channel-force margin, dominated by the first days at flood stage, is the

margin the portfolio cannot insure, and the conflict response concentrates in the paddy-floodplain systems whose agriculture engages river water, not where water-vulnerable crops dominate. One methodological note: the flood channel operates chiefly within the year, the agricultural-damage channel into the following one, so testing the flood channel against the wrong lag will fail.

A.10. Modeled Flooding versus the Flood Catalogue

The natural alternative to a routing model is the Global Flood Database ([Tellman et al., 2021](#)), and the reason to decline it is measurement ([Patel, 2024](#)). Every one of its 147 India events descends from the news-based Dartmouth catalogue, and the error is testable in-sample: small floods are missing non-randomly, detection depends on who lives downstream, and detection varies with cloud cover — which tracks the storm itself. The regression the catalogue invites shows the consequence: a catalogued-event indicator predicts *fewer* recorded conflict events, but the GFD’s own mapped extent carries nothing, a CaMa-built indicator matched to the catalogue’s base rate is significantly *positive*, and the negative appears identically in two independent news-derived conflict datasets — attention displacement within a shared news pipeline, not pacification. The design therefore uses continuous physical quantities throughout and retains the GFD in the validation role below.

A.11. Construction and Validation

Masked runs.—The upstream-only discharge series comes from CaMa-Flood run with the district’s own runoff input removed: grid cells within 50 km of the district receive zero runoff — physically identical to zero local rain — so the routed flow at the district’s main river cell is upstream-origin by construction. Districts that can be masked simultaneously without contaminating one another share simulations (34 runs at the production footprint, each district losing at most 5 percent of upstream input to co-members’ masks). Return-period thresholds are Gumbel fits (L-moments) to the 19 annual maxima of the unmasked baseline at each cell; restart-based event reruns reproduce the baseline bit-exactly.

Is the routing itself credible? Against VIC-RRM — an independent model with different physics and routing, calibrated on Bangladesh gauge observations — CaMa’s daily discharge agrees closely: $r = 0.96$ at the Ganges checkpoint, median $r = 0.90$ across 365 district gauge cells, and median $\kappa = 0.48$ on the margin the design actually uses, which days exceed a cell’s own return level. The models share ERA5-Land forcing, so this validates routing rather than meteorology. The resulting ordering is respected throughout: discharge is validated twice over, inundated extent is biased in level but valid in rank, and modelled flood *duration* is not used.

Definition robustness.—Rebuilding the ensemble with the mask drawn at the paper’s excl.-neighbor footprint instead of 50 km gives instruments correlating 0.94 within district and a flood-stage result of 0.024 (50 km) versus 0.021 (excl.-neighbor) on the common sample. The 50-km construction is the better-identified of the two: excluding districts whose mask removes under 1 percent of upstream input leaves its estimate at 0.020 while the administrative variant loses significance, and the one excl.-neighbor run that failed its volume-conservation gate did so for exactly this reason and is held out.

Validation against satellite flooding.—Over the 147 catalogued GFD events, the model’s inundation extent correlates with observed flooded fractions at 0.41 (Spearman, district level), reaching 0.51 with a probability of detection of 0.90 at unit-catchment resolution on major events. The model is naturalized — no reservoirs or levees — so infrastructure-failure floods like the 2008 Kosi embankment breach are invisible to it by construction: the routed instrument captures rain-driven floods only, precisely the exogeneity property the design wants.

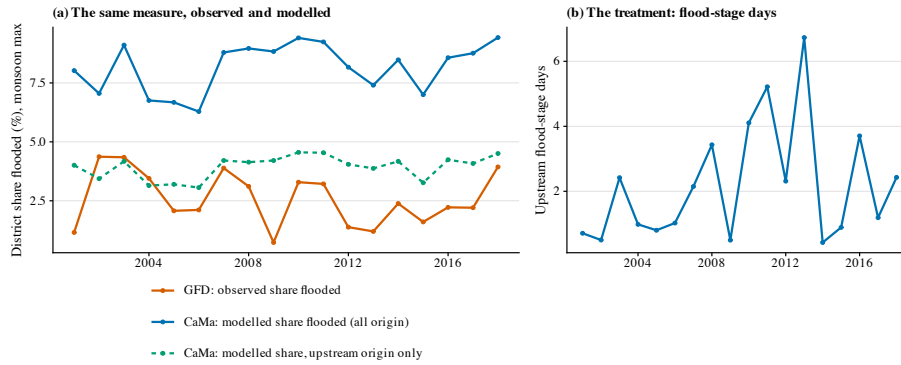


Figure A.8: Routed and observed flooding over time

Notes: Yearly means across the districts carrying both sources (387 districts with a CaMa river cell, an upstream signal and no bifurcation flag). Panel (a) is in levels — the monsoon-maximum percentage of district area flooded — for GFD (satellite-observed, any origin), the CaMa baseline run (modelled, any origin), and the CaMa masked run (modelled, upstream origin only). Panel (b) is the paper’s treatment: days the three-day-mean upstream-origin discharge exceeds the two-year return level, averaged over the same districts. GFD district-years lost to cloud cover, and districts without CaMa floodplain cells, drop from their series’ means.

The flood threshold and its vintage.—

Does the flood threshold define the result?—The return level is what makes a day a flood day. Refitting the levels on the full 37-year routing record rather than the 19 annual maxima the published panel used raises the two-year level by a median 4.5% and cuts mean flood-stage days on the race sample from 2.13 to 1.79, while the two series still correlate at 0.973. Put side by side in the specification that races them, nothing moves: routed flood days read 0.0274 against 0.0274, upstream rainfall 0.775 against 0.782. The ordering Section VI draws on does not depend on where the threshold was placed.

Appendix B. Additional Figures and Tables

Exhibits are grouped by the main-text section they support; each carries its own notes.

B.1. Descriptive Figures

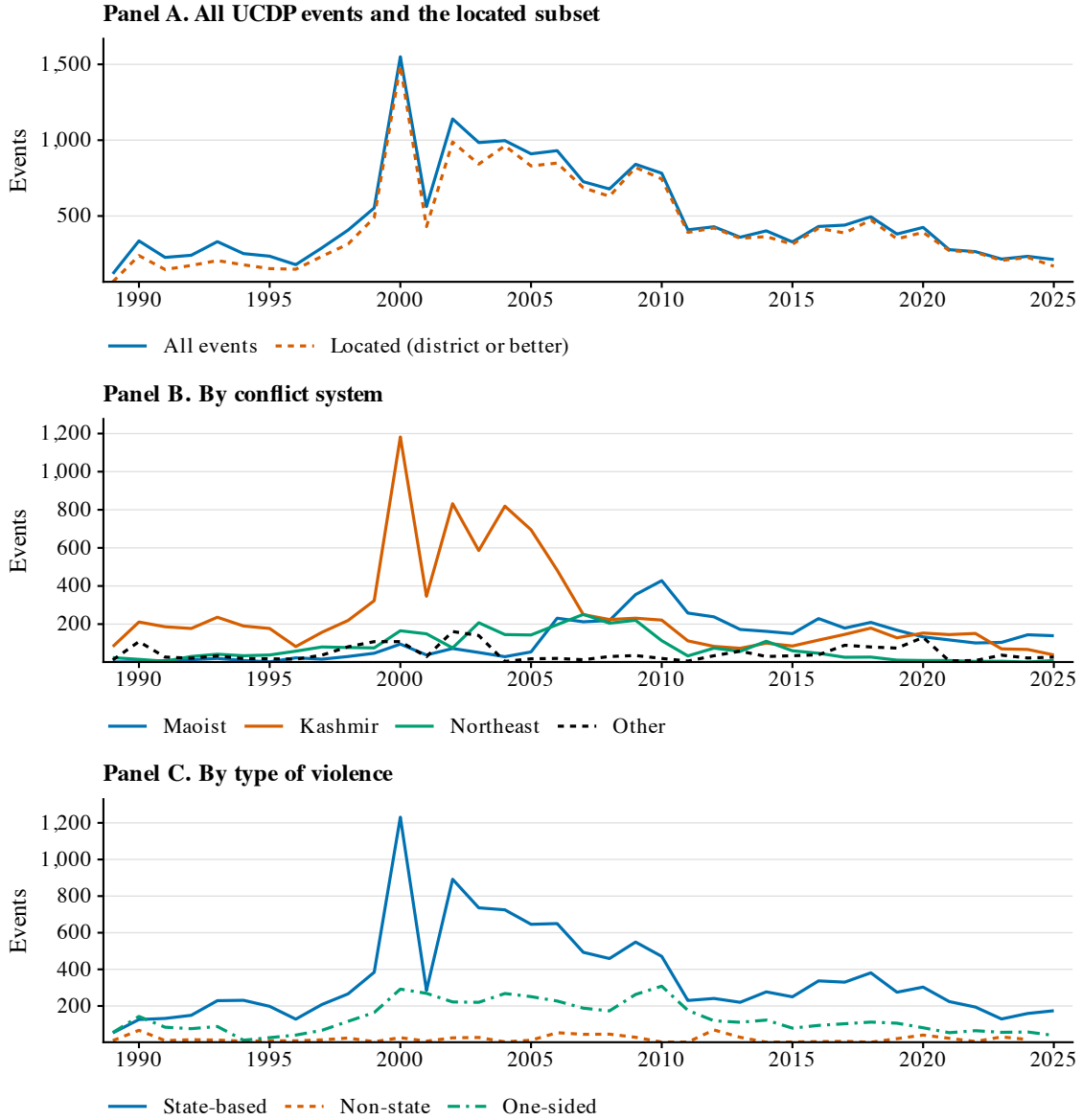


Figure B.1: UCDP conflict events in India, 1989–2025

Notes: Panel A plots all events against the subset UCDP locates to a district or better, the outcome used throughout the paper; the two are drawn on one count axis, and the gap between them is the share of events UCDP could place no more finely than a state. Panel B splits all events by conflict system, assigned from the UCDP conflict name (Maoist, Kashmir, Northeast, and a residual holding India–Pakistan interstate events, communal non-state violence, and unaffiliated one-sided violence). Panel C splits them by UCDP’s type of violence. Series are distinguished by line style as well as colour, so the figure reads in greyscale.

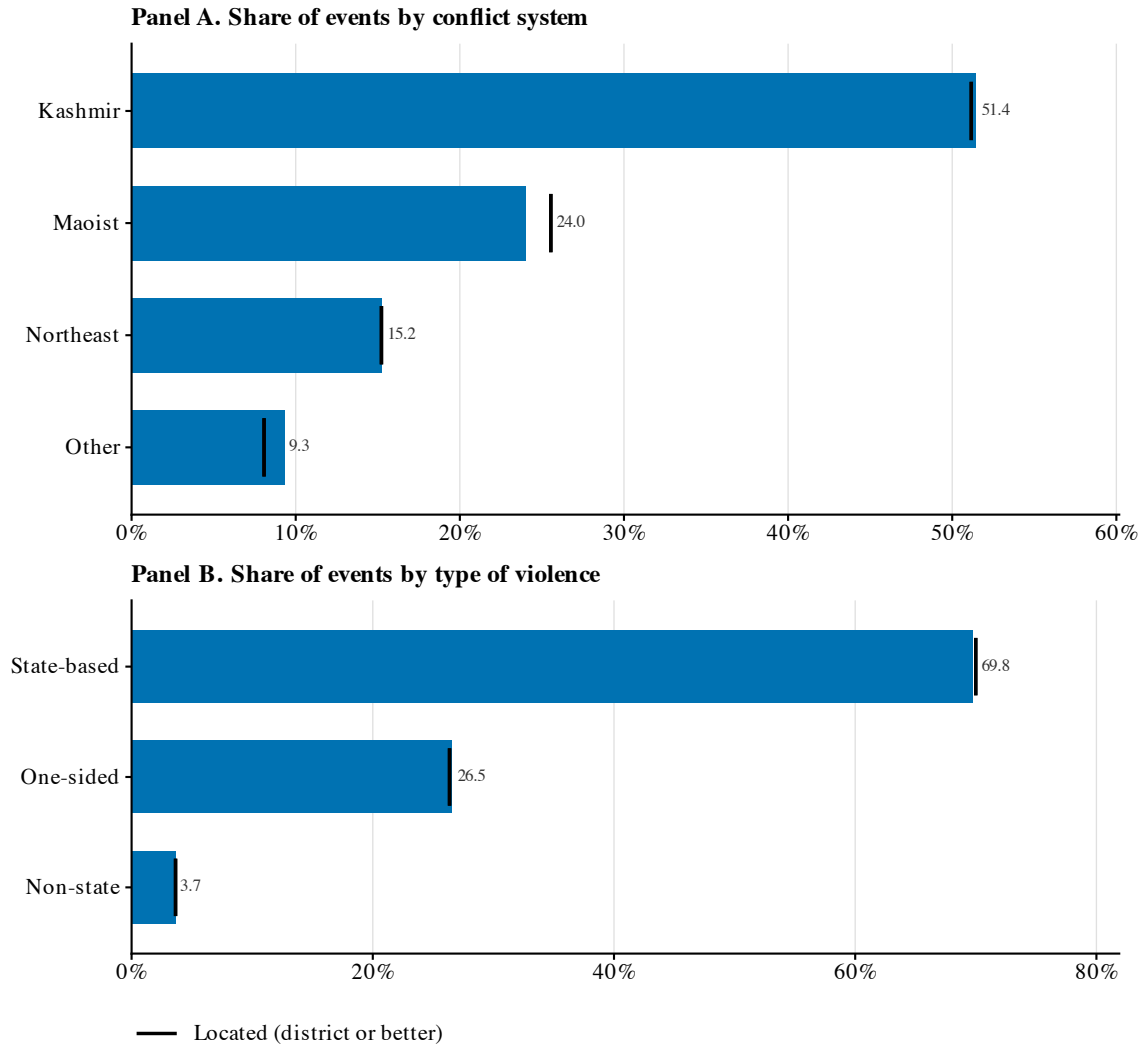


Figure B.2: Composition of UCDP events in India, 1989–2025

Notes: Bars give each category's share of all events over the full record; the tick on each bar gives its share of the located subset the estimation uses. Bars and ticks coincide to within 1.5 percentage points everywhere, which is the point of plotting them together: the precision restriction is not selective on conflict system or on type of violence. That neutrality is about composition only: the same restriction is highly selective across *districts* — state-located events pile onto one representative point per state — which is what drives the Kashmir result. Categories are as in Figure B.1.

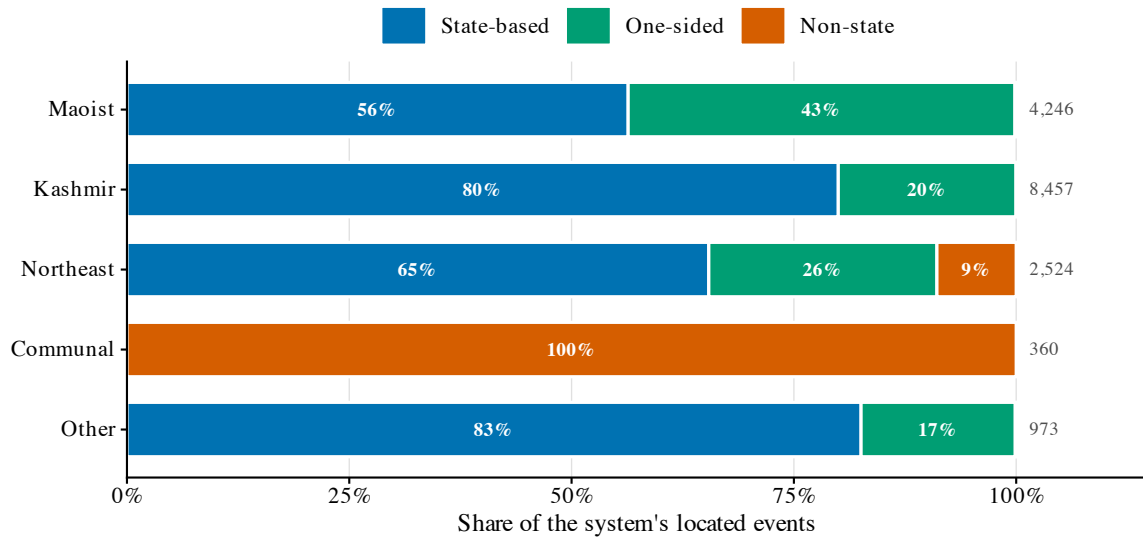


Figure B.3: Mix of violence types within each conflict system, located events, 1990–2025

Notes: Figure B.2 gives each category's share of all events; this figure gives the cross-tabulation, which is what Tables B.3 and 3 condition on. Bars are each system's located events divided among UCDP's three types of violence and sum to 100 percent; the count beside each bar is the events it rests on. Two of the rows are definitional rather than empirical: communal violence is non-state by construction, which is why it contributes no column to Table 3, and the unaffiliated residual is one-sided by construction. Among the three systems the paper estimates, Maoist violence is the most evenly divided (56.3 percent state-based against 43.5 percent one-sided), which is why its pooled count moves least: the two components respond in opposite directions and largely cancel. Kashmir is 80.0 percent state-based and the Northeast 65.4 percent. Restricting to located events changes no cell of this table by more than 2.4 percentage points, so the composition here is not an artifact of the precision rule. Every segment carries its value, so the figure reads in greyscale.

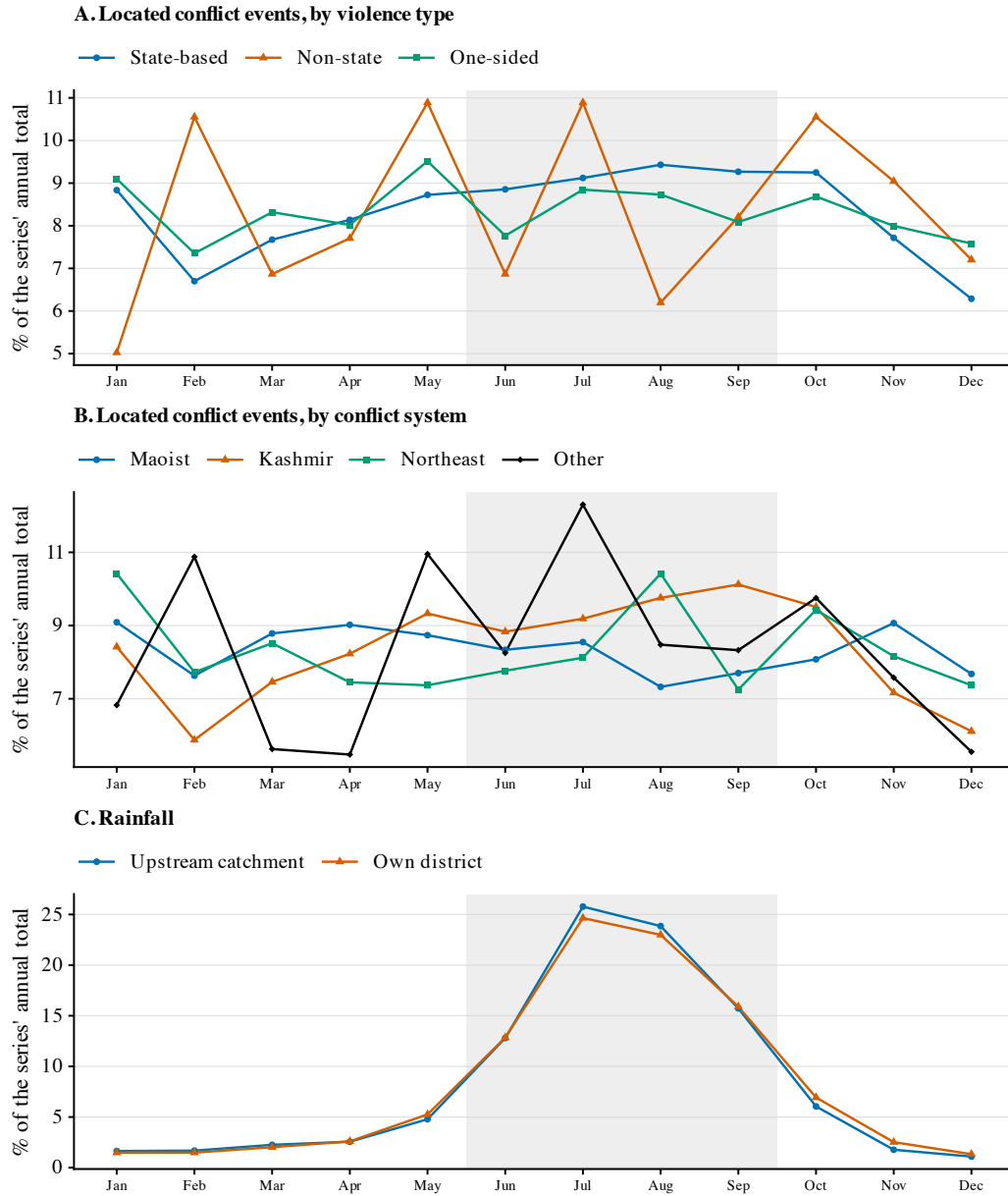


Figure B.4: Monthly composition of conflict and rainfall, 1990–2025

Notes: Each series is normalised to sum to 100 percent across the twelve calendar months, so the panels show *when* each series occurs rather than how large it is. Panels A and B use located UCDP events; violence types are UCDP's state-based, non-state and one-sided categories, and conflict systems are the paper's Maoist, Kashmir and Northeast groupings with everything else pooled into Other. Panel C rebuilds upstream rainfall from the basin-month series through the same crosswalk and the same inverse-distance weight, $1/(1 + d)$, that the annual treatment uses, so the seasonal shape shown is the treatment's own; district monthly shares are computed first and then averaged across districts, so that a few very wet districts cannot set the national profile. The shaded band marks June–September.

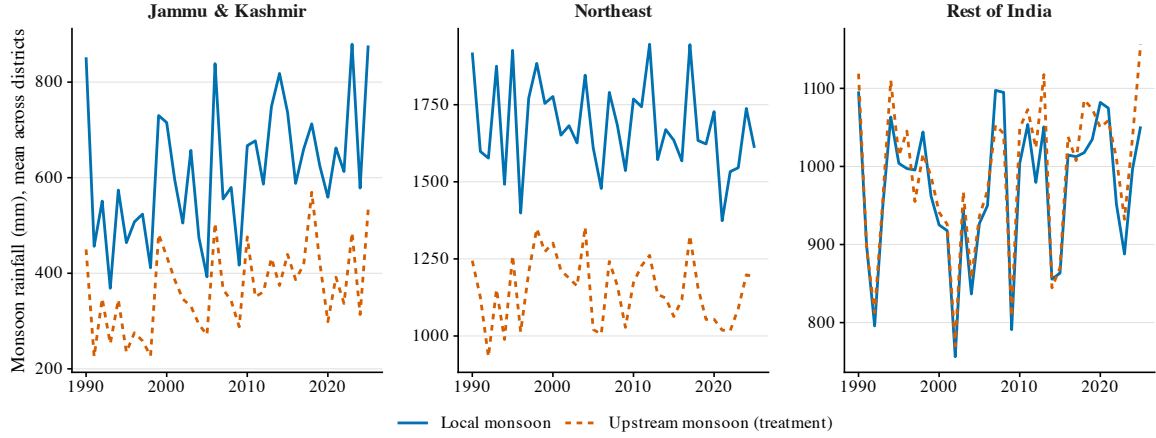


Figure B.5: Monsoon rainfall over time, local and upstream, by region

Notes: Yearly means across the estimation-sample districts of the two rainfall variables the regressions use: the district's own monsoon total and the inverse-distance-weighted monsoon total over its excl.-neighbour upstream catchment. Regions are assigned by state (Jammu & Kashmir; the seven Northeastern states; the rest), and the y-scales differ by region because the levels do.

B.2. Main-Result Robustness

Table B.1: The reduced form under alternative fixed effects, windows, and the downstream placebo

	Upstream ($t - 1$)	Uncertainty	Districts
<i>A. Fixed effects (published window)</i>			
District + year (baseline)	0.87	Conley 0.38	—
State $\times$ year	0.81	SE 0.43, $p = 0.061$	—
Region $\times$ year	1.19	$p = 0.001$	—
State linear trends	1.03	$p = 0.015$	—
<i>B. Sample windows</i>			
2000–2018 (crop window)	1.33	$t = 2.67$	—
2000–2025	1.60	$t = 3.61$	—
1989–2025	0.89	$t = 2.64$	233
<i>C. The downstream twin (annual rainfall)</i>			
Downstream alone, own sample	−1.42	SE 0.36	212
Downstream alone, common sample	−1.84	—	160
Downstream, three-way race	−1.95	—	160
Upstream in the same race	1.04	SE 0.38	160

Notes: Panel A varies the fixed effects on the published specification; state $\times$ year absorbs everything common to a state in a year, leaving districts of the same state draining different catchments. Panel B varies the window; the reduced form needs no crop data. Panel C reports the downstream placebo: the downstream series is built by the same procedure following the flow network in the opposite direction, the two treatments correlate 0.56 after fixed effects, and collinear readings of one river corridor split a common effect — the sign pattern, not the magnitude, is the informative object, and a misassigned upstream catchment would push the downstream term positive. Diagnostics retired from the prose: the lead of annual upstream rainfall is -0.01 ($t \approx 0$); upstream and local monsoon rainfall correlate 0.76 after fixed effects and the local controls are retained throughout; the effective number of independent shocks is 1,339 sub-basins, 39.5 PFAF-5 catchments and 7.64 macro basins, and a shock-level Borusyak–Hull–Jaravel transformation reproduces the point estimate with a first-stage F below 3, so nothing is concluded from it. Standard errors clustered on district unless stated.

Table B.2: Is the monsoon the right window?

	Monsoon	Crop-calendar alternatives		
	(1)	(2)	(3)	(4)
log Upstream Precip., monsoon ($t - 1$)	1.410*** (0.329)		1.481*** (0.332)	1.429*** (0.335)
log Upstream Precip., growing season ($t - 1$)		0.717** (0.323)		
log Upstream Precip., non-monsoon window ($t - 1$)			-0.080 (0.112)	
Growing-season component orthogonal to the monsoon ($t - 1$)				-0.329 (0.291)
Observations	8,280	8,050	8,050	8,050
Number of Districts	230	230	230	230
Mean of dependent variable	1.01	1.02	1.02	1.02
Estimation	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓

Notes: Every column is the specification of Table 2, column 4, on 1990–2025; only the rainfall window changes. *Crop-calendar weights.* MIRCA2000 reports, for each district, the share of its cropped area under crops in each calendar month; these shares are time-invariant. Monthly upstream rainfall is multiplied by those shares and summed, so column 2 weights the whole year by when that district’s own agriculture actually happens rather than by a calendar common to all districts. *The non-monsoon term.* Columns 3 and 4 split the weights at the monsoon boundary and renormalise each half to sum to one, so the non-monsoon regressor carries *only* October–May rainfall, weighted by the share of the district’s cropping falling in those months — broadly the rabi season. That window straddles two calendar years, so it is dated by monsoon year: the term dated t runs from October of year t to May of year $t + 1$, and the lagged term covers the same eight months one year earlier. *Column 4.* The regressor is the residual from projecting the whole-year crop-calendar measure on monsoon rainfall within district and year — the crop-calendar content the monsoon window does not already carry. It is a generated regressor and its standard errors are not corrected for that first step. Current-period terms, weather controls and the spatial block are estimated throughout and not displayed. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table B.4: Where the upstream effect lands: own district against successive rings

	Own only (1)	+ Ring 1 (2)	+ Rings 1 and 2 (3)
<i>Panel A. Upstream monsoon rainfall</i>			
Own upstream	0.775*** (0.265)	0.864** (0.373)	0.888** (0.382)
Ring 1 (adjacent)		-0.257 (0.667)	0.262 (0.765)
Ring 2 (adjacent to adjacent)			-0.807* (0.463)
Total	0.775*** (0.265)	0.607 (0.495)	0.343 (0.511)
Residual corr. with own	—	0.889	0.889, 0.824
<i>Panel B. Routed flood-stage days (CaMa)</i>			
Own upstream	0.017** (0.007)	-0.003 (0.017)	0.012 (0.014)
Ring 1 (adjacent)		0.043 (0.034)	-0.012 (0.030)
Ring 2 (adjacent to adjacent)			0.070* (0.038)
Total	0.017** (0.007)	0.040** (0.020)	0.070** (0.031)
Residual corr. with own	—	0.741	0.741, 0.454
Estimation	Poisson	Poisson	Poisson
District & Year FE	23✓	✓	✓

Table B.3: Which conflicts respond, cut each way

	All conflict (1)	State-based (2)	One-sided (3)	Non-state (4)
<i>Panel A. By type of violence</i>				
log Upstream Precip. ($t - 1$)	1.410*** (0.329)	1.236*** (0.366)	1.340** (0.666)	2.705** (1.208)
log Upstream Precip. (t)	0.022 (0.455)	0.078 (0.472)	-0.387 (0.649)	0.131 (1.379)
log Local Precip. (t)	-0.187 (0.561)	0.002 (0.596)	-1.454* (0.781)	2.605 (1.797)
log Local Precip. ($t - 1$)	0.400 (1.002)	0.801 (1.211)	-0.752 (1.062)	4.406* (2.564)
1 SD of identifying variation (log points) equivalently, a wetter monsoon of (%)	0.11 12	0.11 12	0.11 12	0.11 12
Implied effect of a 1 SD wetter monsoon (%)	17	15	16	35
Observations	8,280	5,616	5,976	1,960
Districts	230	156	166	70
Mean of dependent variable	1.01	0.94	0.44	0.19
	Maoist (5)	Kashmir (6)	Northeast (7)	Communal (8)
<i>Panel B. By conflict system</i>				
log Upstream Precip. ($t - 1$)	0.603 (0.627)	0.663* (0.401)	2.203*** (0.665)	2.115* (1.221)
log Upstream Precip. (t)	-1.296* (0.764)	-0.450 (0.309)	0.082 (1.075)	0.405 (1.527)
log Local Precip. (t)	-1.163 (1.179)	-0.323 (0.433)	0.172 (1.071)	1.705 (2.310)
log Local Precip. ($t - 1$)	-1.603 (1.075)	0.876 (1.091)	-0.230 (0.563)	3.991 (2.702)
1 SD of identifying variation (log points) equivalently, a wetter monsoon of (%)	0.10 10	0.13 14	0.09 10	0.11 12
Implied effect of a 1 SD wetter monsoon (%)	6	9	22	27
Observations	3,535	1,620	1,365	1,488
Districts	101	45	39	62
Mean of dependent variable	0.99	1.30	1.52	0.20
Estimation	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓

Notes: Every cell is a separate Poisson regression on 1990–2025 under the specification of Table 2, column 4: district and year fixed effects, the full weather controls, and the spatial block. Only the outcome changes. Neighbours' conflict in the spatial block is built from the column's own outcome, so each column is conditioned on neighbouring violence of its own kind. Types are UCDP's type of violence, systems are assigned from the UCDP conflict name. Poisson drops districts recording no events of a given kind, which is why the district counts differ down the table. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table B.5: What the spatial specification includes, and what happens if it does not

	log Upstream Precip. ($t - 1$)		Observations
<i>Panel A. Count outcome, fixed-effects Poisson</i>			
As estimated: $\lambda(Wy)_{t-1}$ and $\theta(WX)_t$	1.410***	(0.329)	8,280
+ own temporal lag γy_{t-1} (<i>inconsistent</i>)	1.289***	(0.439)	8,280
<i>Panel B. Linear outcome, $\log(1 + \text{count})$</i>			
Without γy_{t-1}	0.062**	(0.026)	16,884
+ own temporal lag γy_{t-1}	0.060***	(0.023)	16,884
+ contemporaneous spatial lag $\rho(Wy)_t$, by IV	0.086	(0.107)	16,884

Notes: Longest window, district and year fixed effects, Conley (150 km) standard errors in parentheses. The specification carries neighbours' lagged conflict and neighbours' weather but neither the contemporaneous spatial lag nor the district's own temporal lag, and this table gives the reason for each omission and shows what relaxing it does. *The own temporal lag.* A fixed-effects Poisson conditions on each district's own total count over the panel. $y_{i,t-1}$ is a component of that total and carries district i 's fixed effect mechanically, so conditioning no longer removes the incidental parameter and the estimator is inconsistent. The space-time lag $(Wy)_{i,t-1}$ is built from the *neighbours'* outcomes, whose fixed effects are absorbed by their own dummies and never enter i 's conditioning set; it is predetermined in the ordinary sense and needs only no-serial-correlation, not strict exogeneity. That distinction is what makes the combination principled rather than convenient. Row 2 imposes the term anyway: the elasticity is 1.289 against 1.410, so the omission conceals nothing. In the linear arm, where the term is admissible, adding it moves the coefficient from 0.062 to 0.060. Nickell bias is of order $1/T$ with $T = 36$ and is noted rather than solved, as in the source literature. *The contemporaneous spatial lag.* It is simultaneously determined — a district's violence enters its neighbours' regressor while theirs enters its own — so it requires either instruments or maximum likelihood. The conventional Kelejian–Prucha instrument, second-order neighbours' weather, has a first-stage F of 0.526; the resulting estimate is uninformative (last row) and $\hat{\rho}$ is -0.602. The instrument with real first-stage power on this panel — second-order neighbours' *upstream* exposure — is invalid, because catchments nest: it correlates 0.93 with a district's own treatment, so it reaches the outcome directly rather than only through the neighbours. The maximum-likelihood route of Harari and Ferrara (2018) is defined for a linear model of conflict incidence and has no standard counterpart for a fixed-effects Poisson count, which is the outcome here. Both conventional solutions to the simultaneity are therefore closed on the count margin, and the specification restricts to terms that need neither.

Table B.6: Spatial diffusion and the upstream treatment

	Incidence (LPM)				asinh(events)
	Contiguity (1)	River W (2)	Both (3)	Downstream (4)	River W (5)
Upstream rain ($t - 1$)	0.034** (0.017)	0.042** (0.018)	0.030 (0.031)	0.038** (0.018)	0.118*** (0.031)
Own conflict ($t - 1$)	0.176*** (0.025)	0.192*** (0.026)	0.176*** (0.025)	0.197*** (0.026)	0.437*** (0.033)
Neighbours' conflict ($t - 1$), contiguity	0.237*** (0.036)		0.208*** (0.040)		
Upstream districts' conflict ($t - 1$), river W		0.183*** (0.039)	0.057 (0.042)		0.173*** (0.046)
Downstream districts' rain (t)				-0.022 (0.017)	
<i>Panel B: event counts (Poisson), 176 districts</i>					
	Baseline	+ river conflict lag	+ contiguity too		
Upstream rain ($t - 1$)	0.802* (0.440)	1.150*** (0.368)	1.035*** (0.364)		
<i>Panel C: the flood arm — routed flood-stage days (Poisson), 152 districts</i>					
	Baseline	+ river lag	+ queen lag	+ upstr. flood days	
Flood-stage days (t)	0.024*** (0.006)	0.022*** (0.007)	0.023*** (0.007)	0.016* (0.008)	
Flood-stage days ($t - 1$)	0.019*** (0.007)	0.018*** (0.007)	0.017*** (0.006)	0.018*** (0.006)	
Upstream districts' flood days (t)				0.061* (0.035)	

Notes: Time-space recursive models with Durbin-type exogenous lags (Online Appendix B); 2000–2018; district and year fixed effects; standard errors clustered by district in parentheses. These are NOT dynamic spatial Durbin models in the sense of Harari and Ferrara (2018): the contemporaneous spatial lag $\rho(Wy)_t$ is absent throughout, for the reasons Table B.5 sets out. One consequence is interpretive and favourable — with no contemporaneous spatial lag there is no feedback loop, so these coefficients are effects directly and need no direct–indirect decomposition. Columns (1)–(4) are linear probability models of conflict incidence on all districts; column (5) uses asinh of the event count. Column (1) is the contiguity (queen) network and also holds neighbours' weather fixed; columns (2)–(5) use the directed river network, with (3) entering both conflict lags jointly and (4) the downstream direction. Panel B estimates the count model by fixed-effects Poisson with the own conflict lag excluded (strict exogeneity) on a frame without the upstream temperature controls, so its baseline sits below the published headline and comparisons are internal to the panel. Panel C repeats the exercise for the routed flood-stage days of the flood arm, local weather held throughout and the published no-controls form reproduced first; its final column adds upstream districts' own flood days, whose post-fixed-effects correlation with own days is 0.51 — a partial overlap with interpretable shares, not the near-collinear splits flagged in the text. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

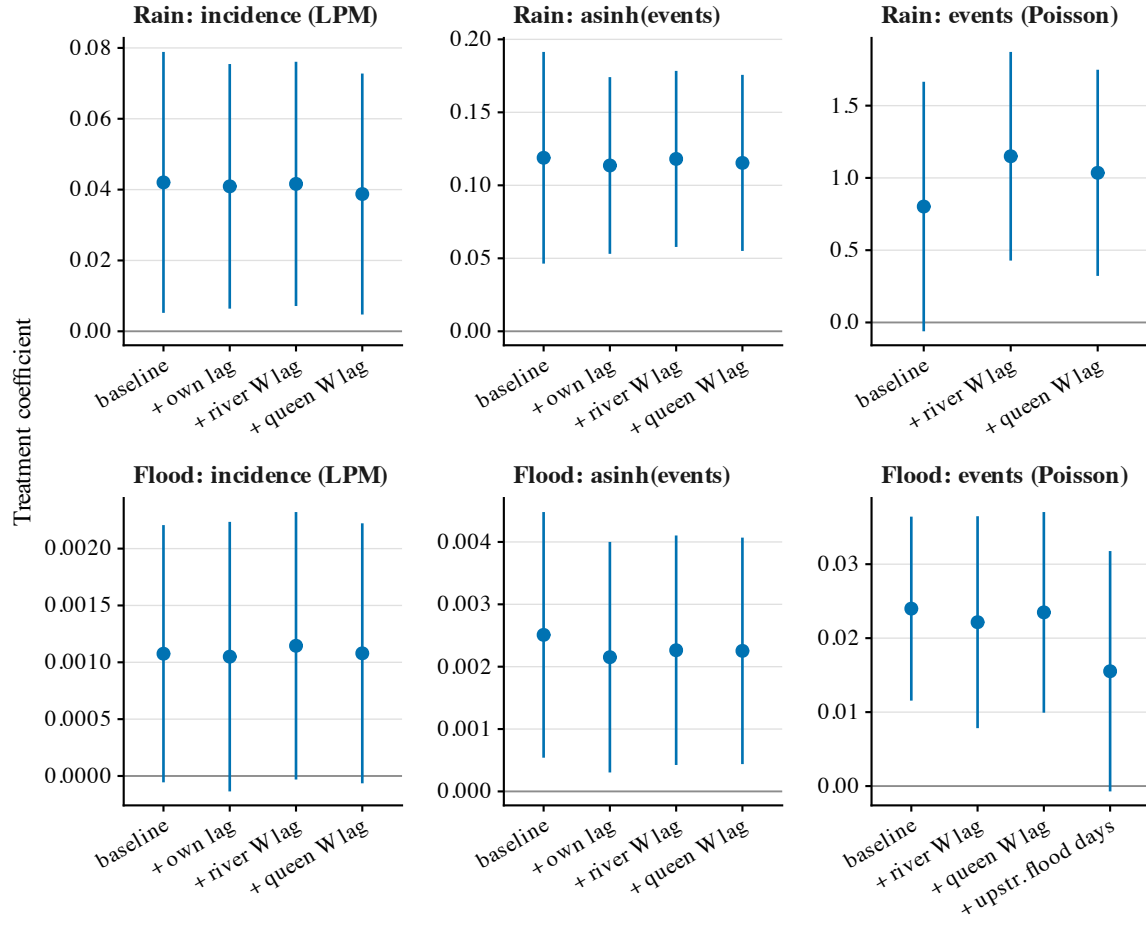


Figure B.6: The stability ladders: treatment coefficients as spatial controls enter, with 95 percent confidence intervals (district-clustered)

Notes: Top row: lagged upstream rainfall. Bottom row: contemporaneous routed flood-stage days. Left and centre columns are the linear models of Table B.6; the right column is the Poisson count model, where for rainfall the coefficient rises under diffusion controls and for flood days the final rung adds upstream districts' own flood days (the same-wave control, post-FE correlation 0.51). Poisson ladders omit the own-conflict lag throughout (strict exogeneity); the rainfall Poisson frame lacks the upstream temperature controls, so its baseline is not the published headline.

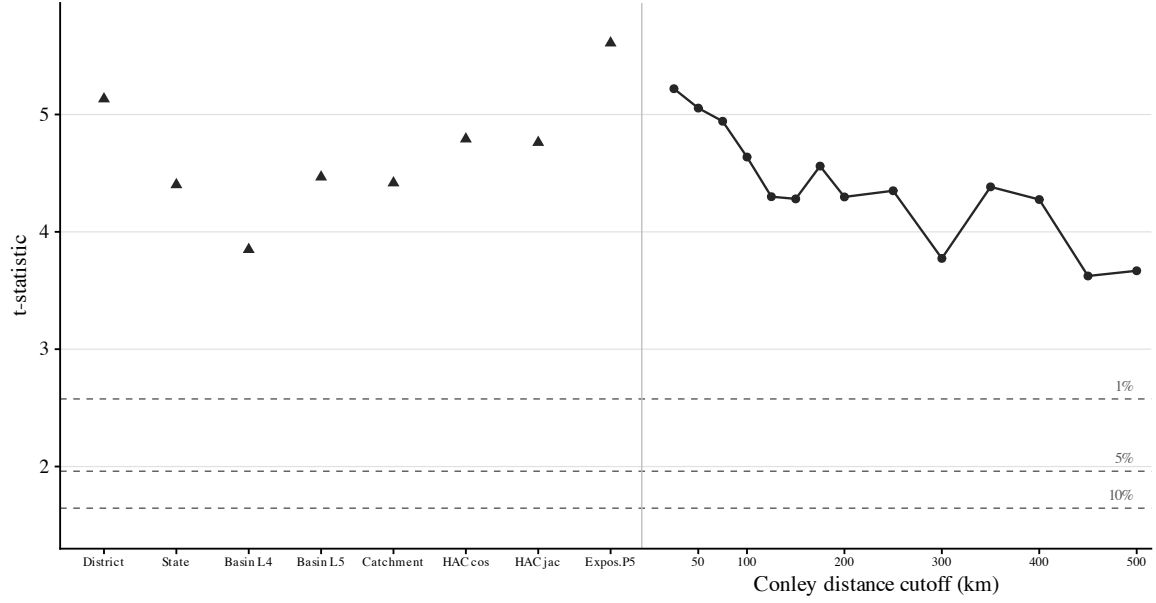


Figure B.7: The headline test statistic under alternative variance estimators

Notes: Every point is the same regression — the specification of Table 2, column 4, estimated on 1990–2025 — re-priced under a different variance estimator, so the figure varies the estimator and nothing else. The point estimate is 1.41 throughout and t is that estimate divided by the standard error in question. The connected curve gives Conley spatial-HAC statistics at distance cutoffs from 25 to 500 kilometres; the grid stops at 500 because beyond it the kernel encompasses most of the country and no longer describes local dependence. The eight triangles at the left have no distance cutoff and are shown at their own labelled positions. Five cluster the residuals — on districts, states, PFAF level-4 and level-5 macro basins, and the catchment grouping built from the treatment itself in this section. The last three do not cluster at all: ‘HAC cos’ and ‘HAC jac’ are spatial-HAC estimators whose kernel is the cosine, respectively Jaccard, similarity of districts’ upstream-weight vectors, so two districts fed by the same basins are treated as dependent however far apart they lie; ‘Expos. P5’ is the exposure-robust estimator of [Adão et al. \(2019\)](#), which treats the PFAF level-5 basin shocks rather than the districts as the sampling unit. The exposure-robust estimator computed at the finest basin level is deliberately omitted: with more than a thousand basin shocks its asymptotics do not bite and it returns a standard error of 0.130, far tighter than anything else here, which would imply a precision the design does not have. Dashed lines mark the two-sided normal critical values at the 10, 5 and 1 percent levels.

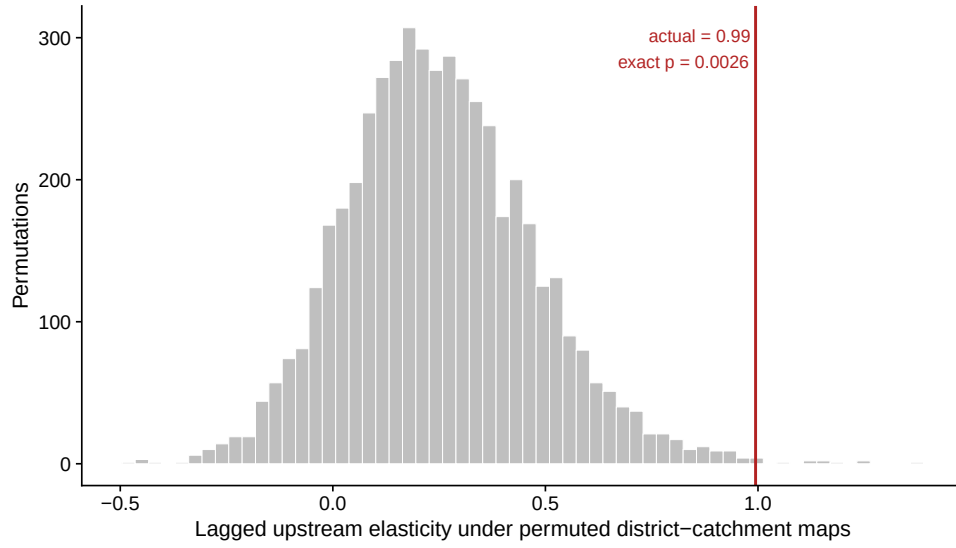


Figure B.8: Randomization inference

Notes: Distribution of the lagged upstream elasticity across 5,000 within-state permutations of the district-catchment assignment; the red line is the estimate under the true assignment. The placebo distribution sits above zero because within-state permutation preserves the regional monsoon, which makes the test conservative.

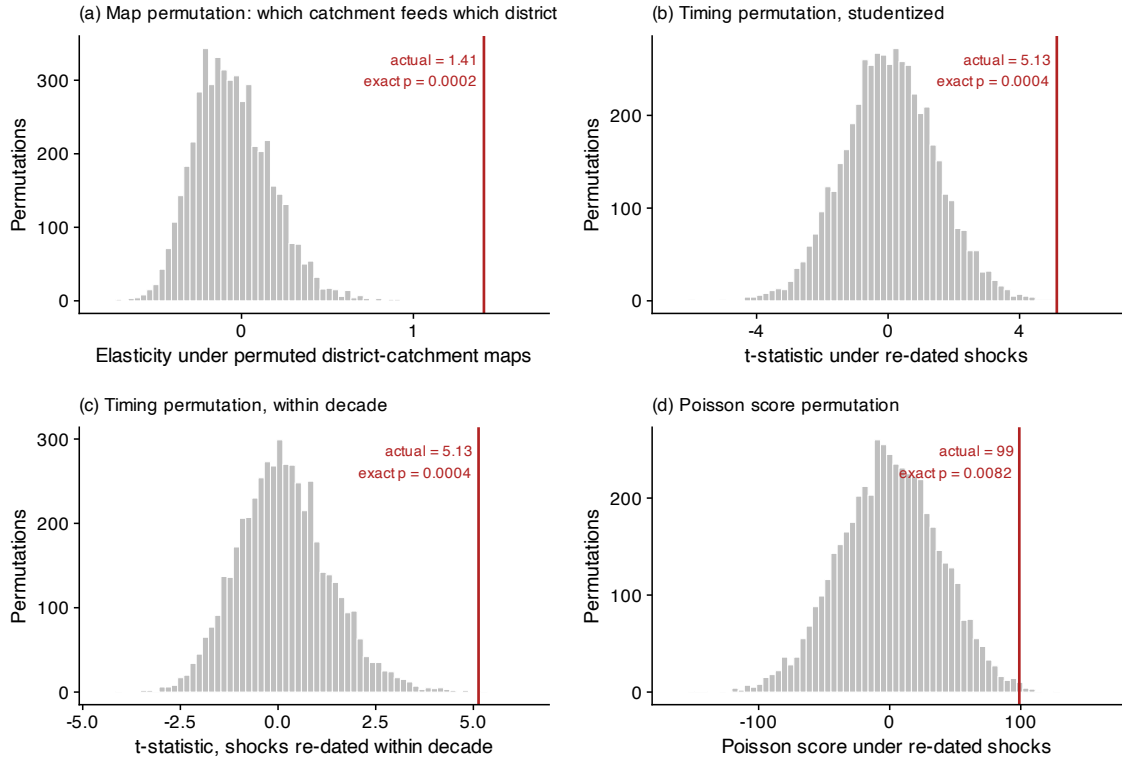


Figure B.9: Four randomization tests of the same estimate

Notes: Every panel re-estimates the specification of Table 2, column 4 on 1990–2025 under 5,000 permutations; the grey histogram is the null distribution, the vertical rule the observed statistic. Panel (a) permutes the district–catchment map within states, holding time fixed. Panels (b)–(d) permute the year labels of the whole cross-section jointly, holding the map fixed: one permutation per draw applied to every district, so the spatial dependence of the shock is preserved and only the shock–outcome alignment breaks. The outcome and the spatial terms built from it are never re-dated. Panel (c) restricts each permutation to stay within its decade, which preserves trend and relaxes the exchangeability the free permutation assumes. Panel (d) uses the score statistic implied by the Poisson likelihood. Permuting the upstream rainfall term alone rather than the whole forcing block gives $p = 0.0044$ on the studentized statistic; that narrower version leaves the treatment’s own temperature control and the local weather at their true dates, so the regressors are no longer mutually aligned.

Table B.7: Flooding at migrant origins and conflict at destinations

Origin definition	FloodMig (t)	Cluster SE	Perm. p (coef.)	Perm. p (t)
Origin state (D-02)	-0.059	(0.028)	0.109	0.150
Origin district (D-11)	0.017	(0.009)	0.532	0.244
Hybrid	0.016	(0.014)	0.613	0.526

Notes: Poisson fixed-effects, 2002–2018, the destination’s own routed flood days and local weather included in every column. FloodMig is the migration-share-weighted mean of origin flood days, with shares from the 2001 Census. D-02 gives origin states only; D-11 gives origin districts within a state. No definition produces an effect surviving the year-block permutation, and the district coefficient is a precise zero among destinations that never flood.

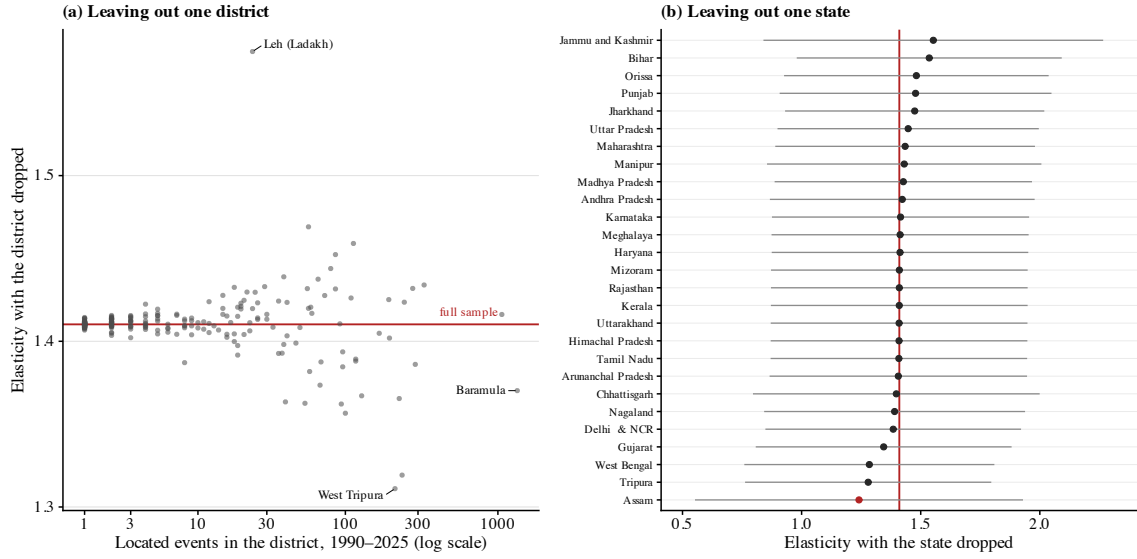


Figure B.10: Leave-one-out robustness, by district and by state

Notes: Each point re-estimates the specification of Table 2, column 4 on 1990–2025 with one district (panel a) or one state (panel b) removed; the red line is the full-sample estimate 1.41. Panel (a) plots the leave-it-out estimate against the dropped district’s own count of located events, so the vertical spread shows influence and the horizontal position shows concentration; the three labelled districts are the two extremes and the most violent. Panel (b) shows 95 percent district-clustered intervals, states ordered by the resulting estimate, with Assam — the most influential — marked. Districts are dropped from the frame before estimation, so the spatial block is rebuilt on the reduced sample rather than carried over.

B.3. Mechanism

Table B.8: Kharif crop margins: extensive and intensive

	Rice (1)	Cotton (2)	Bajra (3)	Maize (4)	Groundnut (5)	Pigeonpea (6)
<i>Area, extensive: any planting (0/1)</i>						
log Upstream Precip. (<i>t</i>)	-0.074*** (0.014)	-0.077*** (0.024)	-0.067*** (0.022)	-0.134*** (0.022)	-0.071*** (0.026)	-0.060** (0.024)
log Downstream Precip. (<i>t</i>)	0.044** (0.018)	0.014 (0.023)	-0.000 (0.022)	0.090*** (0.023)	0.015 (0.028)	0.023 (0.026)
Observations	9,744	9,744	9,744	9,744	9,744	9,744
<i>Area, intensive: log hectares, growing district-years</i>						
log Upstream Precip. (<i>t</i>)	0.019 (0.076)	0.198 (0.128)	-0.023 (0.114)	-0.317*** (0.081)	0.035 (0.121)	-0.173* (0.094)
log Downstream Precip. (<i>t</i>)	0.057 (0.075)	-0.143 (0.107)	0.110 (0.110)	0.162 (0.100)	0.381*** (0.115)	0.117 (0.106)
Observations	7,248	4,161	5,649	8,062	5,836	7,524
<i>Production, extensive: any output (0/1)</i>						
log Upstream Precip. (<i>t</i>)	-0.076*** (0.014)	-0.063*** (0.024)	-0.060*** (0.023)	-0.162*** (0.027)	-0.077*** (0.026)	-0.044* (0.024)
log Downstream Precip. (<i>t</i>)	0.044** (0.018)	0.009 (0.023)	0.008 (0.023)	0.099*** (0.025)	0.018 (0.028)	0.035 (0.026)
Observations	9,744	9,744	9,744	9,744	9,744	9,744
<i>Production, intensive: log tonnes, producing district-years</i>						
log Upstream Precip. (<i>t</i>)	0.258*** (0.091)	0.463*** (0.168)	0.097 (0.137)	-0.205** (0.090)	0.126 (0.133)	-0.108 (0.099)
log Downstream Precip. (<i>t</i>)	0.013 (0.088)	-0.236* (0.137)	0.235 (0.144)	0.239** (0.101)	0.430*** (0.111)	0.018 (0.116)
Observations	7,240	4,035	5,599	7,878	5,772	7,413

Notes: Every cell is one regression on the specification of Table B.14's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Extensive rows are linear probability models on an indicator for any planting (or any output) among district-years where the crop is observed; intensive rows are logs on the district-years that grow (produce) it. Columns are the six largest crops by the season's planted area, grown in at least forty districts. The season is the monsoon's own: rain falls on the standing crop, so the extensive rows measure plantings prevented or destroyed. ***, **, and * denote 1%, 5%, and 10%.

Table B.9: Rabi crop margins: extensive and intensive

	Wheat (1)	Chickpea (2)	Rape & mustard (3)	Jowar (4)	Maize (5)	Barley (6)
<i>Area, extensive: any planting (0/1)</i>						
log Upstream Precip. (<i>t</i>)	-0.008 (0.014)	-0.020 (0.020)	0.009 (0.024)	0.028 (0.017)	-0.000 (0.021)	0.011 (0.017)
log Downstream Precip. (<i>t</i>)	-0.004 (0.011)	-0.032 (0.023)	0.068*** (0.020)	-0.057*** (0.020)	-0.043** (0.022)	0.001 (0.014)
Observations	9,613	9,613	9,613	9,613	9,613	9,613
<i>Area, intensive: log hectares, growing district-years</i>						
log Upstream Precip. (<i>t</i>)	0.220*** (0.068)	-0.141 (0.096)	-0.104 (0.078)	-0.284 (0.230)	-0.100 (0.219)	0.005 (0.078)
log Downstream Precip. (<i>t</i>)	-0.116* (0.059)	0.143 (0.103)	-0.060 (0.074)	0.287 (0.231)	0.708*** (0.238)	0.178 (0.110)
Observations	8,805	8,166	8,051	1,470	3,377	5,109
<i>Production, extensive: any output (0/1)</i>						
log Upstream Precip. (<i>t</i>)	-0.016 (0.014)	-0.019 (0.020)	0.019 (0.023)	0.030* (0.017)	-0.000 (0.022)	0.021 (0.016)
log Downstream Precip. (<i>t</i>)	-0.004 (0.011)	-0.036 (0.023)	0.059*** (0.020)	-0.059*** (0.020)	-0.043* (0.022)	-0.007 (0.014)
Observations	9,613	9,613	9,613	9,613	9,613	9,613
<i>Production, intensive: log tonnes, producing district-years</i>						
log Upstream Precip. (<i>t</i>)	0.242*** (0.079)	-0.163 (0.103)	-0.137 (0.091)	-0.246 (0.237)	-0.223 (0.223)	0.035 (0.082)
log Downstream Precip. (<i>t</i>)	-0.226*** (0.066)	0.116 (0.107)	-0.105 (0.079)	0.293 (0.248)	0.831*** (0.248)	0.135 (0.123)
Observations	8,788	8,028	7,988	1,462	3,368	5,007

Notes: Every cell is one regression on the specification of Table B.14's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Extensive rows are linear probability models on an indicator for any planting (or any output) among district-years where the crop is observed; intensive rows are logs on the district-years that grow (produce) it. Columns are the six largest crops by the season's planted area, grown in at least forty districts. The monsoon precedes rabi sowing, so the contemporaneous term is rain on the soil moisture the winter crop is sown into. Rabi area is concentrated — wheat alone is three-fifths of it — so the smaller columns carry wide intervals. ***, **, and * denote 1%, 5%, and 10%.

Table B.10: Kharif crop prices and the value of the harvest

	Rice (1)	Cotton (2)	Bajra (3)	Maize (4)	Groundnut (5)	Pigeonpea (6)
<i>Farm-harvest price, annual (log)</i>						
log Upstream Precip. (<i>t</i>)	0.014 (0.023)	0.026 (0.034)	-0.049* (0.026)	-0.057*** (0.022)	-0.067** (0.027)	0.002 (0.024)
log Downstream Precip. (<i>t</i>)	-0.060*** (0.020)	0.054* (0.032)	-0.001 (0.023)	0.022 (0.024)	0.011 (0.022)	0.033 (0.023)
Observations	5,900	2,273	4,278	5,698	4,853	4,641
<i>Revenue: price $\times$ production (log)</i>						
log Upstream Precip. (<i>t</i>)	0.448*** (0.110)	0.003 (0.342)	0.008 (0.186)	-0.180 (0.118)	0.111 (0.193)	0.098 (0.161)
log Downstream Precip. (<i>t</i>)	-0.116 (0.096)	-0.186 (0.223)	0.214 (0.159)	0.366** (0.142)	0.329** (0.144)	0.174 (0.218)
Observations	2,832	1,197	2,435	3,143	2,702	2,900

Notes: Every cell is one regression on the specification of Table B.14's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Prices are the DLD farm-harvest series, which is annual — it is not reported by season — so the revenue rows pair the annual price with kharif production: roughly the value of the harvest, and gross revenue rather than profit, since input costs are not observed. Because $\log(pq) = \log p + \log q$, the revenue coefficient equals the sum of the price and production responses on its own sample; it will not equal the sum of the rows printed above it or in Table B.8, which are estimated on the larger samples their own margins allow. The row's contribution is the standard error of that sum, which the covariance of the parts makes unreadable from the separate margins. Both blocks condition on producing district-years with a price quote, so they are intensive margins. ***, **, and * denote 1%, 5%, and 10%.

Table B.11: Kharif crop margins on the conflict-estimation sample

	Rice (1)	Cotton (2)	Bajra (3)	Maize (4)	Groundnut (5)	Pigeonpea (6)
<i>Area, extensive: any planting (0/1)</i>						
log Upstream Precip. (<i>t</i>)	-0.057*** (0.020)	-0.022 (0.036)	-0.037 (0.033)	-0.146*** (0.035)	-0.041 (0.040)	-0.019 (0.034)
log Downstream Precip. (<i>t</i>)	0.069*** (0.024)	-0.054** (0.027)	-0.023 (0.032)	0.104*** (0.035)	-0.038 (0.039)	0.018 (0.038)
Observations	4,553	4,553	4,553	4,553	4,553	4,553
<i>Area, intensive: log hectares, growing district-years</i>						
log Upstream Precip. (<i>t</i>)	-0.016 (0.100)	0.029 (0.192)	-0.096 (0.182)	-0.436*** (0.141)	0.043 (0.202)	-0.144 (0.111)
log Downstream Precip. (<i>t</i>)	0.203* (0.108)	-0.113 (0.129)	0.159 (0.158)	0.162 (0.127)	0.427*** (0.163)	0.135 (0.111)
Observations	3,051	1,977	2,218	3,580	2,253	3,608
<i>Production, extensive: any output (0/1)</i>						
log Upstream Precip. (<i>t</i>)	-0.061*** (0.020)	-0.008 (0.036)	-0.031 (0.034)	-0.172*** (0.038)	-0.034 (0.042)	-0.010 (0.033)
log Downstream Precip. (<i>t</i>)	0.070*** (0.024)	-0.056** (0.028)	-0.022 (0.033)	0.097*** (0.037)	-0.030 (0.040)	0.022 (0.037)
Observations	4,553	4,553	4,553	4,553	4,553	4,553
<i>Production, intensive: log tonnes, producing district-years</i>						
log Upstream Precip. (<i>t</i>)	0.300** (0.132)	0.609** (0.263)	0.062 (0.196)	-0.198 (0.165)	0.256 (0.213)	0.001 (0.130)
log Downstream Precip. (<i>t</i>)	0.172 (0.128)	-0.108 (0.189)	0.150 (0.183)	0.276** (0.140)	0.496*** (0.151)	0.103 (0.127)
Observations	3,047	1,924	2,204	3,477	2,220	3,575

Notes: Every cell is one regression on the specification of Table B.14's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Extensive rows are linear probability models on an indicator for any planting (or any output) among district-years where the crop is observed; intensive rows are logs on the district-years that grow (produce) it. Columns are the six largest crops by the season's planted area, grown in at least forty districts. The sample is restricted to the districts the conflict Poisson retains — at least one located UCDP event over 1990–2025; columns are kept identical to the full-sample table for comparability. ***, **, and * denote 1%, 5%, and 10%.

Table B.12: Rabi crop margins on the conflict-estimation sample

	Wheat (1)	Chickpea (2)	Rape & mustard (3)	Jowar (4)	Maize (5)	Barley (6)
<i>Area, extensive: any planting (0/1)</i>						
log Upstream Precip. (<i>t</i>)	-0.065*** (0.021)	-0.047 (0.032)	-0.041 (0.028)	0.021 (0.024)	-0.021 (0.037)	0.005 (0.029)
log Downstream Precip. (<i>t</i>)	0.006 (0.013)	-0.054 (0.034)	0.038 (0.024)	-0.040 (0.028)	-0.034 (0.029)	0.005 (0.023)
Observations	4,586	4,586	4,586	4,586	4,586	4,586
<i>Area, intensive: log hectares, growing district-years</i>						
log Upstream Precip. (<i>t</i>)	0.194 (0.124)	-0.279** (0.138)	-0.327*** (0.113)	-0.744 (0.638)	-0.322 (0.340)	0.184 (0.130)
log Downstream Precip. (<i>t</i>)	-0.203** (0.084)	0.012 (0.119)	-0.030 (0.087)	0.505 (0.602)	1.463*** (0.330)	0.019 (0.133)
Observations	4,248	3,916	4,056	487	1,714	2,269
<i>Production, extensive: any output (0/1)</i>						
log Upstream Precip. (<i>t</i>)	-0.074*** (0.022)	-0.045 (0.032)	-0.048* (0.028)	0.021 (0.024)	-0.024 (0.038)	0.017 (0.028)
log Downstream Precip. (<i>t</i>)	0.006 (0.013)	-0.058* (0.035)	0.036 (0.024)	-0.040 (0.028)	-0.032 (0.029)	-0.008 (0.023)
Observations	4,586	4,586	4,586	4,586	4,586	4,586
<i>Production, intensive: log tonnes, producing district-years</i>						
log Upstream Precip. (<i>t</i>)	0.150 (0.133)	-0.365** (0.148)	-0.457*** (0.132)	-0.292 (0.644)	-0.407 (0.348)	0.159 (0.133)
log Downstream Precip. (<i>t</i>)	-0.295*** (0.094)	0.010 (0.125)	-0.104 (0.092)	0.843 (0.597)	1.606*** (0.333)	-0.016 (0.150)
Observations	4,239	3,873	4,040	485	1,710	2,234

Notes: Every cell is one regression on the specification of Table B.14's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Extensive rows are linear probability models on an indicator for any planting (or any output) among district-years where the crop is observed; intensive rows are logs on the district-years that grow (produce) it. Columns are the six largest crops by the season's planted area, grown in at least forty districts. The sample is restricted to the districts the conflict Poisson retains — at least one located UCDP event over 1990–2025; columns are kept identical to the full-sample table for comparability. Rabi jowar falls to 33 growing districts here, below the forty-district floor the full-sample selection imposes; the column is retained for comparability and should be read accordingly. ***, **, and * denote 1%, 5%, and 10%.

Table B.13: Kharif crop prices and the value of the harvest on the conflict-estimation sample

	Rice (1)	Cotton (2)	Bajra (3)	Maize (4)	Groundnut (5)	Pigeonpea (6)
<i>Farm-harvest price, annual (log)</i>						
log Upstream Precip. (<i>t</i>)	0.033 (0.032)	0.015 (0.043)	-0.034 (0.041)	-0.078** (0.033)	-0.058 (0.038)	0.078** (0.031)
log Downstream Precip. (<i>t</i>)	-0.066*** (0.024)	0.116** (0.048)	-0.024 (0.035)	-0.022 (0.029)	0.003 (0.032)	0.050 (0.036)
Observations	3,390	1,243	1,820	2,935	2,201	2,322
<i>Revenue: price × production (log)</i>						
log Upstream Precip. (<i>t</i>)	0.413*** (0.143)	-0.063 (0.490)	-0.267 (0.313)	-0.108 (0.204)	0.046 (0.318)	0.249 (0.211)
log Downstream Precip. (<i>t</i>)	-0.025 (0.111)	-0.078 (0.297)	0.216 (0.215)	0.317* (0.171)	0.538*** (0.164)	0.507* (0.284)
Observations	1,293	648	879	1,396	995	1,421

Notes: Every cell is one regression on the specification of Table B.14's agricultural rows — upstream, local, neighbour and downstream monsoon rainfall, each contemporaneous and lagged, temperature controls, district and year fixed effects, standard errors clustered on district; the contemporaneous upstream and downstream terms are shown. Prices are the DLD farm-harvest series, which is annual — it is not reported by season — so the revenue rows pair the annual price with kharif production: roughly the value of the harvest, and gross revenue rather than profit, since input costs are not observed. Because $\log(pq) = \log p + \log q$, the revenue coefficient equals the sum of the price and production responses on its own sample; it will not equal the sum of the rows printed above it or in Table B.8, which are estimated on the larger samples their own margins allow. The row's contribution is the standard error of that sum, which the covariance of the parts makes unreadable from the separate margins. Both blocks condition on producing district-years with a price quote, so they are intensive margins. ***, **, and * denote 1%, 5%, and 10%. The sample is restricted to the districts the conflict Poisson retains — at least one located UCDP event over 1990–2025; columns are kept identical to the full-sample table for comparability.

Table B.14: Four sources of water in one regression

	Each outcome's own sample				Conflict-estimation districts			
	Upstream (1)	Local (2)	Neighbour (3)	Downstream (4)	Upstream (5)	Local (6)	Neighbour (7)	Downstream (8)
<i>A. Conflict</i>								
Conflict events (Poisson) ($t - 1$)	1.162*** (0.363)	0.882 (0.812)	-1.651 (1.136)	-0.916*** (0.324)	1.162*** (0.363)	0.882 (0.812)	-1.651 (1.136)	-0.916*** (0.324)
<i>B. District agriculture, prices and wages</i>								
Kharif planted area (t)	-0.127* (0.067)	-0.031 (0.177)	0.253 (0.193)	0.143* (0.076)	-0.355*** (0.120)	0.040 (0.259)	0.318 (0.324)	0.344*** (0.087)
Fixed-weight quantity index ($t - 1$)	0.056 (0.073)	-0.161 (0.169)	0.288 (0.199)	0.036 (0.063)	-0.074 (0.139)	-0.009 (0.301)	0.099 (0.366)	0.057 (0.101)
Implicit price index ($t - 1$)	-0.030 (0.021)	0.023 (0.039)	-0.064 (0.041)	0.052*** (0.018)	-0.039 (0.025)	0.023 (0.067)	-0.054 (0.074)	0.032 (0.028)
Fixed-weight price index (t)	-0.003 (0.017)	0.033 (0.035)	-0.071 (0.044)	0.009 (0.016)	0.011 (0.027)	0.043 (0.062)	-0.063 (0.079)	0.013 (0.022)
Upstream districts' output (t)	0.317*** (0.049)	-0.013 (0.105)	0.084 (0.129)	-0.017 (0.038)	0.245*** (0.067)	0.073 (0.153)	0.066 (0.188)	-0.082* (0.049)
Real agricultural wage ($t - 1$)	-0.135*** (0.034)	0.011 (0.059)	0.005 (0.076)	0.103*** (0.032)	-0.124** (0.049)	-0.069 (0.086)	0.169 (0.117)	0.068 (0.041)
<i>C. IHDS households</i>								
Farm income (asinh) (t)	-0.788 (1.658)	-2.655 (1.942)	2.880 (2.531)	2.535** (1.113)	4.969** (2.381)	-4.120 (2.730)	-0.461 (2.811)	1.378 (1.469)
Non-farm income (asinh) ($t - 1$)	-3.264** (1.316)	1.101 (1.807)	0.857 (2.708)	0.127 (0.882)	-4.765* (2.443)	2.307 (2.812)	2.019 (3.162)	-0.645 (1.373)
Receives any remittance ($t - 1$)	0.126** (0.057)	-0.007 (0.109)	-0.198 (0.138)	-0.066 (0.042)	0.072 (0.108)	0.203 (0.162)	-0.387* (0.205)	-0.058 (0.077)
Consumption (log) (t)	-0.575*** (0.179)	0.634** (0.252)	-0.282 (0.290)	0.047 (0.140)	0.146 (0.319)	-0.277 (0.355)	0.103 (0.315)	-0.255 (0.195)
Districts, conflict row			210				210	
District & Year (or Wave) FE			✓				✓	

Notes: Every row is one regression carrying all four rainfall measures at once, each entered contemporaneously and lagged, with upstream and local temperatures and their lags, district and year (or survey-wave) fixed effects, and standard errors clustered on district. Each row reports one horizon, marked against its label: ($t - 1$) for the lagged term and (t) for the contemporaneous one. Crop quantities and the prices they set are read contemporaneously, since a crop is harvested in the year the rain falls; conflict, wages and the slower household margins are read at $t - 1$. Upstream is the inverse-distance-weighted catchment draining into the district excluding the district and its neighbours; downstream mirrors it exactly, on the catchment the district drains into; neighbour is the row-normalised queen-contiguity average of local rainfall. Columns 1–4 estimate each outcome on whatever districts it supports; columns 5–8 restrict every outcome to the 210 districts the conflict Poisson retains, so the comparison across outcome families holds geography fixed. The crop aggregates respond at the extensive margin — plantings pushed to zero in district-years where the crop is marginal — which Tables B.8 and B.9 decompose by season; the response among surviving plantings is flat to positive. Local and neighbour rainfall correlate 0.964 after fixed effects and carry variance inflation factors of 14.0 and 16.4, so neither is separately identified; upstream and downstream carry 2.50 and 2.26.

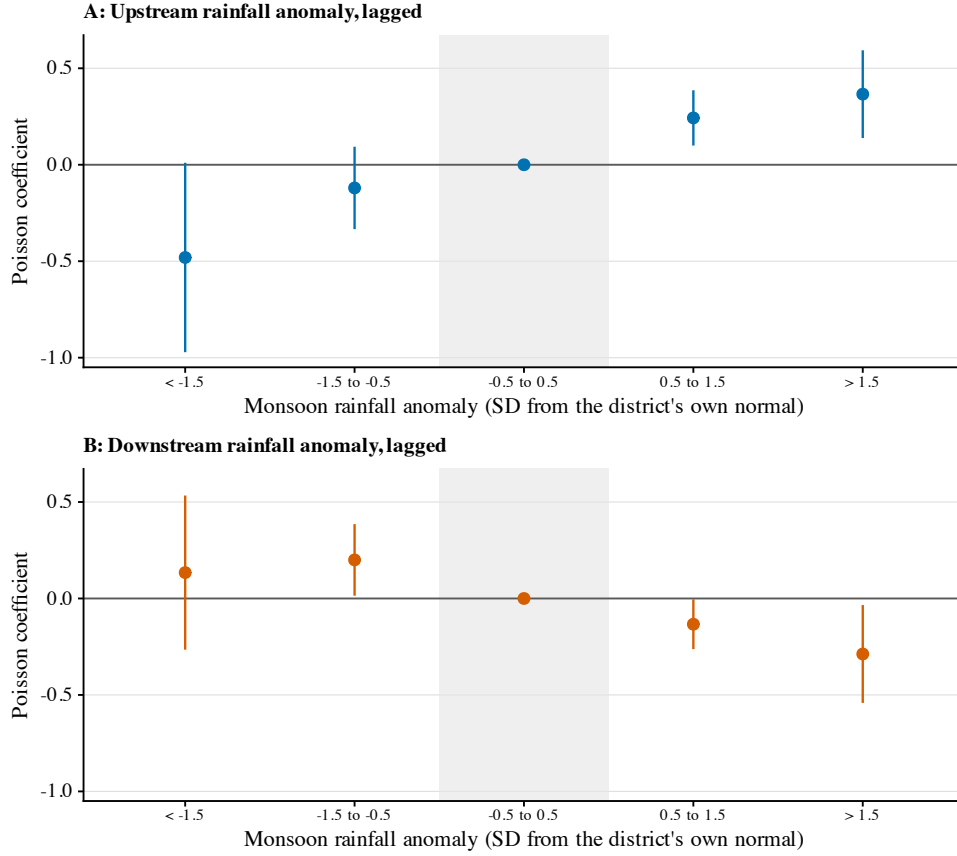


Figure B.11: The dose–response in upstream and downstream rainfall anomalies

Notes: One fixed-effects Poisson regression carrying both sources as five-bin functions of their own standardised monsoon anomaly, with the middle bin (-0.5 to 0.5 standard deviations, shaded) omitted, so every coefficient reads against a normal year. Bins rather than a polynomial: a quadratic imposes a single turning point and would conceal a one-sided response, which is the shape in question. Both terms are lagged, as in the headline. Controls are the local rainfall anomaly and the upstream, downstream and local temperature anomalies, each current and lagged, with district and year effects; standard errors are clustered on district and bars are 95 percent intervals. The control set is the anomaly counterpart of Table B.14 rather than that table’s right-hand side exactly — it carries no neighbour block — so the two are estimated on the same sample and the same sources but are not nested. 7,560 district-years over 210 districts. Both panels share a vertical scale, since the comparison the figure exists to make is one of shape *and* magnitude.

Table B.15: Conflict and the mechanism outcomes in rainfall anomalies

	Conflict events (1)	Planted area (2)	Crops planted (3)	Quantity index (4)	Implicit price (5)	Real wage (6)	Nominal wage (7)
<i>Upstream anomaly, contemporaneous (t)</i>							
< -1.5 SD	0.190 (0.209)	-0.022 (0.062)	0.164*** (0.056)	-0.093 (0.069)	0.006 (0.016)	-0.029 (0.025)	-0.058*** (0.021)
-1.5 to -0.5	-0.071 (0.123)	0.102*** (0.024)	0.091*** (0.027)	0.060** (0.028)	0.010 (0.007)	-0.003 (0.011)	-0.010 (0.009)
0.5 to 1.5	0.183** (0.087)	-0.007 (0.030)	-0.082*** (0.026)	0.017 (0.031)	0.001 (0.007)	-0.037*** (0.014)	-0.019 (0.013)
> 1.5 SD	0.376** (0.156)	0.018 (0.047)	-0.105*** (0.038)	-0.026 (0.049)	0.020 (0.013)	-0.058** (0.023)	-0.027 (0.018)
<i>Upstream anomaly, lagged ($t - 1$)</i>							
< -1.5 SD	-0.581*** (0.222)	0.053 (0.072)	-0.002 (0.053)	0.097 (0.074)	0.011 (0.015)	0.035 (0.023)	-0.009 (0.018)
-1.5 to -0.5	-0.151 (0.110)	0.031 (0.023)	0.034 (0.025)	0.012 (0.027)	0.011 (0.007)	0.010 (0.011)	-0.003 (0.009)
0.5 to 1.5	0.249*** (0.079)	0.054* (0.028)	-0.046* (0.025)	0.038 (0.030)	-0.012 (0.008)	-0.025* (0.014)	-0.015 (0.012)
> 1.5 SD	0.307** (0.128)	0.101** (0.047)	-0.078* (0.045)	0.040 (0.052)	0.005 (0.012)	-0.068*** (0.024)	-0.051** (0.020)
<i>Downstream anomaly, contemporaneous (t)</i>							
< -1.5 SD	0.126 (0.205)	-0.161*** (0.039)	-0.156** (0.062)	-0.151*** (0.043)	0.026* (0.014)	-0.032 (0.022)	-0.001 (0.020)
-1.5 to -0.5	0.312** (0.132)	-0.110*** (0.020)	-0.073*** (0.026)	-0.130*** (0.025)	0.034*** (0.007)	-0.013 (0.013)	0.009 (0.011)
0.5 to 1.5	0.030 (0.129)	0.039* (0.024)	0.034* (0.021)	0.060** (0.026)	0.009 (0.007)	0.009 (0.012)	-0.000 (0.010)
> 1.5 SD	-0.038 (0.126)	0.225*** (0.051)	0.131*** (0.039)	0.236*** (0.052)	0.037*** (0.011)	0.025 (0.021)	0.001 (0.017)
<i>Downstream anomaly, lagged ($t - 1$)</i>							
< -1.5 SD	0.179 (0.208)	0.029 (0.040)	-0.056 (0.049)	0.033 (0.043)	-0.017 (0.014)	-0.028 (0.024)	-0.010 (0.020)
-1.5 to -0.5	0.241** (0.111)	-0.036* (0.022)	-0.014 (0.025)	-0.025 (0.024)	0.009 (0.007)	-0.020 (0.013)	-0.008 (0.012)
0.5 to 1.5	-0.107 (0.080)	0.027 (0.022)	-0.051** (0.024)	0.046* (0.026)	0.017*** (0.007)	0.015 (0.013)	0.007 (0.011)
> 1.5 SD	-0.266** (0.123)	0.167*** (0.051)	0.061 (0.048)	0.144** (0.056)	0.027** (0.013)	0.035 (0.024)	0.017 (0.021)
Estimation	Poisson	OLS	OLS	OLS	OLS	OLS	OLS
Observations	7,560	9,471	9,744	9,471	6,734	4,498	4,498
Districts	210	402	403	402	307	324	324

Notes: Every column is one regression in which upstream, downstream and local monsoon rainfall enter as unrestricted bin functions of each basin-set's own standardised anomaly — the deviation from the district's long-run mean divided by its own standard deviation — at both horizons jointly, with contemporaneous and lagged temperature anomalies as controls and district and year fixed effects; standard errors are clustered on district. The normal bin (−0.5 to 0.5 SD) is omitted, so every coefficient reads against a normal year; local rainfall's bins are controls and are not shown. Mechanism outcomes use the six-crop constructions of Table B.14 on each outcome's own sample, 1997–2022, by OLS; conflict is the fixed-effects Poisson on located events over 1990–2025, the paper's estimation sample. Crops planted counts how many of the six kharif crops have positive area, on district-years observing all six. ***, **, and * denote 1%, 5%, and 10%.

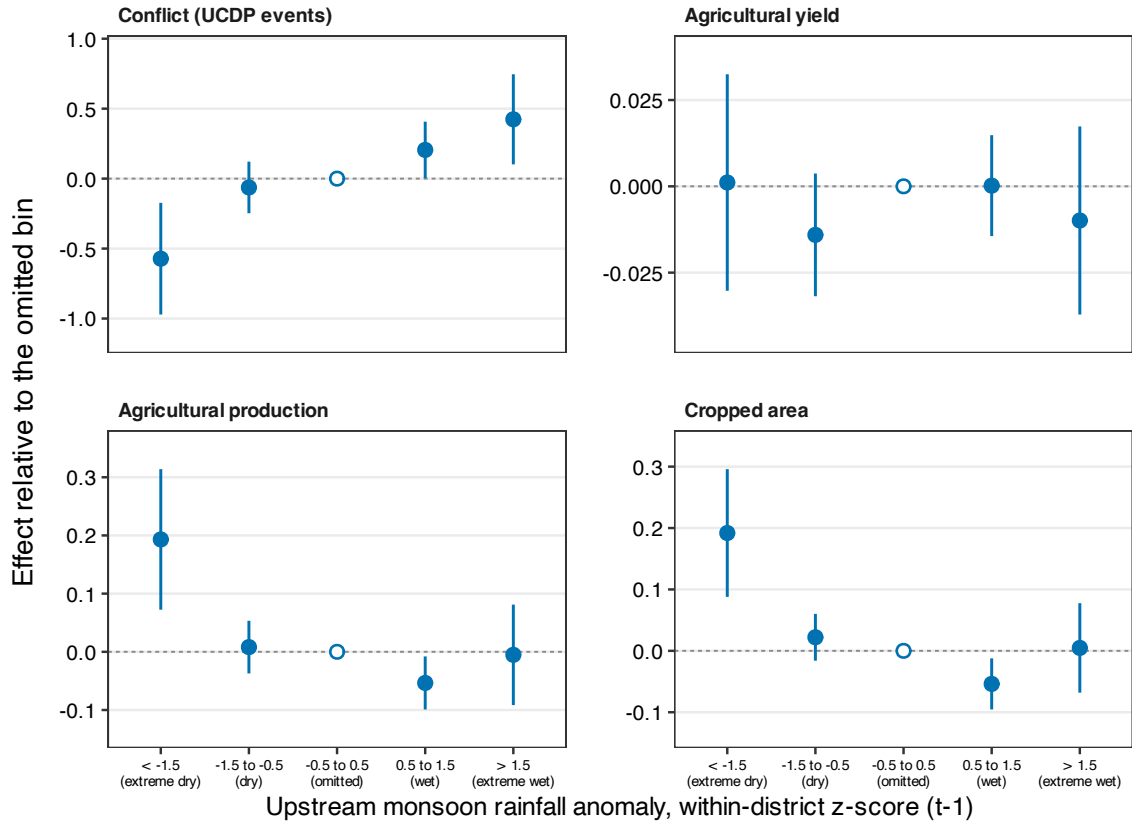


Figure B.12: Conflict and agriculture across the rainfall anomaly

Notes: Points are coefficients on bins of the lagged upstream rainfall anomaly, relative to the omitted -0.5 to 0.5 bin, with 95 percent cluster-by-district intervals. The anomaly is a within-district z -score computed on each district's own annual history. The conflict panel is Poisson on located events over 1990–2025 under the specification of Table 2, column 4 — the same sample of 8,280 district-years and 230 districts as the headline. The agricultural panels are OLS on log outcomes over the years those series cover. Bins rather than a quadratic, because a quadratic imposes one turning point and lets the tails drive the curvature through leverage.

Table B.16: The wage response is a multi-year adjustment

	Real wage (1)	Nominal wage (2)
log Upstream Precip. $(t - 1)$	-0.076** (0.036)	-0.060** (0.027)
log Upstream Precip. $(t - 2)$	-0.038* (0.020)	-0.010 (0.019)
log Upstream Precip. $(t - 3)$	-0.036* (0.022)	-0.031 (0.019)
log Upstream Precip. $(t - 4)$	0.008 (0.022)	-0.019 (0.019)
log Upstream Precip. $(t - 5)$	-0.011 (0.023)	-0.017 (0.020)
Cumulative, $\sum_{k=1}^5$	-0.152* (0.080)	-0.138* (0.071)
Observations	4,674	4,674
Districts	324	324

Notes: Five annual lags of log upstream monsoon rainfall entered together, with local, neighbour and downstream rainfall and the temperature controls as in Table 6, district and year fixed effects, and standard errors clustered on district. The cumulative row is the sum of the five lags with the standard error of that sum. Table 6 reports the single lagged term, which attributes to one year a response that arrives over three: the first three lags each carry part of it and the fourth and fifth carry none. The response is therefore weaker than the single-lag coefficient implies, and it is not drift alone: individual years carry it, which two smoothly co-moving series would not produce. The distributional test the wage series invites — female field labour is disproportionately landless — cannot be run: the female series collapses after 2004 and concentrates in five states, so a female-minus-male gap reports on the sample, not on incidence. ***, **, and * denote 1%, 5%, and 10%.

Table B.17: Labour supply and earnings after an upstream shock: NSS agricultural margins, by fieldwork quarter

	Any ag. wage work (1)	Days, ag. wage work (2)	Ag. wage earnings (3)	All wage earnings (4)
<i>All households</i>				<i>n</i> = 668,427
log Upstream Precip. (<i>t</i>)	+0.0120* (0.0070)	+0.0763* (0.0393)	+0.0939** (0.0438)	-0.0224 (0.0746)
log Upstream Precip. (<i>t</i> − 1)	+0.0051 (0.0083)	+0.0409 (0.0477)	+0.1076** (0.0514)	+0.1411 (0.0989)
<i>Interviewed Jul–Sep, during the monsoon</i>				<i>n</i> = 167,909
log Upstream Precip. (<i>t</i>)	+0.0016 (0.0127)	+0.0308 (0.0692)	+0.0153 (0.0812)	-0.1351 (0.1208)
log Upstream Precip. (<i>t</i> − 1)	-0.0147 (0.0141)	-0.0864 (0.0824)	-0.0366 (0.0891)	-0.0787 (0.1553)
<i>Interviewed Oct–Dec, kharif harvest (+2 months)</i>				<i>n</i> = 168,030
log Upstream Precip. (<i>t</i>)	+0.0132 (0.0120)	+0.0708 (0.0702)	+0.1034 (0.0755)	+0.0580 (0.1362)
log Upstream Precip. (<i>t</i> − 1)	-0.0035 (0.0152)	-0.0436 (0.0888)	+0.0639 (0.0962)	+0.2812* (0.1611)
<i>Interviewed Jan–Mar (+5 months)</i>				<i>n</i> = 165,919
log Upstream Precip. (<i>t</i>)	+0.0054 (0.0104)	+0.0242 (0.0651)	+0.0520 (0.0691)	-0.2219* (0.1262)
log Upstream Precip. (<i>t</i> − 1)	+0.0084 (0.0132)	+0.0850 (0.0776)	+0.1205 (0.0845)	+0.0221 (0.1604)
<i>Interviewed Apr–Jun, following year (+8 months)</i>				<i>n</i> = 166,561
log Upstream Precip. (<i>t</i>)	+0.0266*** (0.0101)	+0.1693*** (0.0618)	+0.1974*** (0.0667)	+0.1851 (0.1249)
log Upstream Precip. (<i>t</i> − 1)	+0.0301** (0.0132)	+0.2075*** (0.0742)	+0.2834*** (0.0825)	+0.3219** (0.1611)

Notes: NSS Employment–Unemployment rounds 61, 66 and 68 (2004–05, 2009–10, 2011–12), working-age individuals, on the four sources of water — upstream, local, neighbouring and downstream rainfall — with temperatures, each at the contemporaneous and lagged monsoon, district and round fixed effects, individual age, sex and education, and standard errors clustered on district. Each round follows exactly one monsoon — 2004, 2009 and 2011 — which is the contemporaneous term. THE REFERENCE PERIOD IS ONE WEEK: days run 0–7. Earnings are wage and salary receipts for that same week in asinh, which keeps the structural zeros — most people do no wage work of a given kind in a given week, and the margin that moves is whether they do any — but prices the jump from zero in rupees, so the coefficients read as sign and significance rather than as elasticities. NSS allocates households to fieldwork quarters BY DESIGN, so the panels compare households drawn to be alike rather than households that selected their timing. The quarters sit at different distances from the SAME monsoon, and the panel headings give that distance: July–September is surveyed during it and April–June eight months after it ended, in the following calendar year — every panel is post-treatment. Estimates without the demographic controls are close to these and are in the replication output. ***, **, and * denote 1%, 5%, and 10%. A coefficient of 0.15 on days is about eight days a year if the response were constant across the year, which the panels show it is not. The response is absent in the first two quarters and concentrated in the last, which is the calendar Section VI reports for conflict. Table B.18 reports the non-farm margins on the same specification.

Table B.18: The non-farm margin after an upstream shock: NSS, by fieldwork quarter

	Casual non-farm			Self-employment		Regular salaried
	Any work (1)	Days (2)	Earnings (3)	Any (4)	Days (5)	Earnings (6)
<i>All households</i>						$n = 668,427$
log Upstream Precip. (t)	-0.0070 (0.0087)	-0.0302 (0.0568)	-0.0521 (0.0596)	+0.0103 (0.0186)	+0.1357 (0.1188)	-0.0648 (0.0631)
log Upstream Precip. ($t - 1$)	-0.0358*** (0.0095)	-0.2042*** (0.0606)	-0.2356*** (0.0672)	+0.0186 (0.0208)	+0.1389 (0.1353)	+0.2645*** (0.0709)
<i>Interviewed Jul–Sep, during the monsoon</i>						$n = 167,909$
log Upstream Precip. (t)	-0.0162 (0.0136)	-0.0611 (0.0864)	-0.1172 (0.0876)	+0.0088 (0.0260)	+0.1039 (0.1656)	-0.0491 (0.0992)
log Upstream Precip. ($t - 1$)	-0.0208 (0.0128)	-0.1076 (0.0790)	-0.1048 (0.0886)	+0.0494 (0.0327)	+0.3848* (0.2056)	+0.0529 (0.1130)
<i>Interviewed Oct–Dec, kharif harvest (+2 months)</i>						$n = 168,030$
log Upstream Precip. (t)	-0.0049 (0.0117)	-0.0369 (0.0754)	-0.0296 (0.0843)	+0.0064 (0.0275)	+0.1329 (0.1818)	-0.0067 (0.0946)
log Upstream Precip. ($t - 1$)	-0.0330** (0.0134)	-0.1500* (0.0837)	-0.2084** (0.0952)	-0.0081 (0.0298)	-0.0270 (0.1946)	+0.4215*** (0.1222)
<i>Interviewed Jan–Mar (+5 months)</i>						$n = 165,919$
log Upstream Precip. (t)	-0.0170 (0.0131)	-0.0955 (0.0842)	-0.1350 (0.0924)	+0.0543** (0.0247)	+0.3775** (0.1492)	-0.1347 (0.0990)
log Upstream Precip. ($t - 1$)	-0.0487*** (0.0150)	-0.2935*** (0.0959)	-0.3507*** (0.1050)	+0.0264 (0.0304)	+0.1858 (0.1931)	+0.2470** (0.1094)
<i>Interviewed Apr–Jun, following year (+8 months)</i>						$n = 166,561$
log Upstream Precip. (t)	+0.0102 (0.0116)	+0.0709 (0.0732)	+0.0735 (0.0832)	-0.0248 (0.0262)	-0.0503 (0.1707)	-0.0856 (0.1002)
log Upstream Precip. ($t - 1$)	-0.0410** (0.0176)	-0.2654** (0.1077)	-0.2817** (0.1234)	+0.0098 (0.0320)	+0.0323 (0.2146)	+0.3207** (0.1242)

Notes: NSS Employment–Unemployment rounds 61, 66 and 68 (2004–05, 2009–10, 2011–12), working-age individuals, on the four sources of water — upstream, local, neighbouring and downstream rainfall — with temperatures, each at the contemporaneous and lagged monsoon, district and round fixed effects, individual age, sex and education, and standard errors clustered on district. Each round follows exactly one monsoon — 2004, 2009 and 2011 — which is the contemporaneous term. THE REFERENCE PERIOD IS ONE WEEK: days run 0–7. Earnings are wage and salary receipts for that same week in asinh, which keeps the structural zeros — most people do no wage work of a given kind in a given week, and the margin that moves is whether they do any — but prices the jump from zero in rupees, so the coefficients read as sign and significance rather than as elasticities. NSS allocates households to fieldwork quarters BY DESIGN, so the panels compare households drawn to be alike rather than households that selected their timing. The quarters sit at different distances from the SAME monsoon, and the panel headings give that distance: July–September is surveyed during it and April–June eight months after it ended, in the following calendar year — every panel is post-treatment. Estimates without the demographic controls are close to these and are in the replication output. ***, **, and * denote 1%, 5%, and 10%. This table is the non-farm counterpart to Table B.17, which reports the agricultural margins on the same specification and sample. Read together they are the two sides of the household’s response: agricultural wage work expands while the casual non-farm option contracts, so the shock is not absorbed by reallocation into non-farm work. Self-employment days are unconditional and include own-cultivation, whose EARNINGS the Employment–Unemployment schedule does not collect — the total in Table B.17 is therefore total WAGE earnings, not total income.

Table B.19: The four sources on the Economic Census, three rounds

	Upstream (1)	Local (2)	Neighbour (3)	Downstream (4)
Establishments	0.440*** (0.133)	-0.216 (0.368)	-0.354 (0.474)	0.021 (0.170)
Non-farm employment	0.442*** (0.139)	0.114 (0.377)	-0.369 (0.498)	0.012 (0.177)
Manufacturing employment	0.177 (0.206)	0.170 (0.597)	-0.001 (0.785)	-0.516* (0.272)
Services employment	0.488*** (0.133)	0.112 (0.342)	-0.510 (0.457)	0.093 (0.167)
Observations			1,278	
Districts			426	
District & Round FE			✓	

Notes: The specification of Table B.14 on the Economic Census, whose rounds give three within-district differences against the household survey's one. Each round's treatment is the mean over the years between it and the previous round, and the lagged term is the mean over the window before that. The 1990 round is excluded: its lag window begins in 1981, before the rainfall panel, so every 1990 observation would lose its lagged treatment. The panel is balanced across the three remaining rounds. Upstream rainfall raises establishment counts and non-farm employment, which is the entry margin Section VI reads as diversification out of damaged agriculture. Downstream rainfall does nothing here, so the upstream-downstream mirror documented in Table B.14 is a feature of the agricultural margin rather than of non-farm entry.

Table B.20: Nightlights and the four sources of water, VIIRS

	Mean light (1)	Total light (2)
log Upstream Precip. (t)	-0.063 (0.044)	-0.065 (0.043)
log Upstream Precip. ($t - 1$)	0.031 (0.051)	0.024 (0.052)
log Local Precip. (t)	0.048 (0.062)	0.049 (0.061)
log Local Precip. ($t - 1$)	-0.004 (0.066)	-0.008 (0.064)
log Downstream Precip. (t)	0.205** (0.081)	0.202** (0.079)
log Downstream Precip. ($t - 1$)	0.195** (0.089)	0.197** (0.088)
Observations	3,834	3,834
Number of Districts	426	426
Estimation	OLS	OLS
District & Year FE	✓	✓

Notes: District-year panel, 2013 – 2021 — the VIIRS record. The outcome is the log of district mean radiance in column 1 and of total radiance in column 2. Each column enters all four sources of water under the specification of Table 2: upstream, local and downstream monsoon rainfall, current and lagged, with their temperature controls and the spatial block is deliberately ABSENT here. Neighbours' local rainfall correlates 0.96 with the district's own after fixed effects, so with both in the equation neither is identified; dropping the neighbour terms costs that control and makes the local coefficient readable, which is the point of the column. VIIRS is reported alone because it and DMSP disagree on the sign of the downstream term, and VIIRS is the calibrated instrument: it carries on-board calibration, does not saturate, and resolves roughly 500 metres, whereas DMSP is a six-bit sensor top-coded at 63 whose inter-satellite drift moves precisely the within-district across-year variation this design uses. The cost is the window: VIIRS begins in 2012, so these columns cannot speak to the earlier decades the rest of the paper covers. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table B.21: Is the downstream effect a dam-release effect?

<i>A. Downstream rainfall interacted with dam presence</i>			
	Downstream rain, no dam (1)	× dam upstream (2)	Implied effect with a dam (3)
Conflict events (Poisson)	-1.794*** (0.445)	1.398*** (0.538)	-0.396 —
Kharif cereal area	0.266** (0.117)	0.007 (0.134)	0.273 —
Kharif cereal production	0.383** (0.173)	-0.065 (0.183)	0.318 —
Real agricultural wage	0.090** (0.039)	0.017 (0.034)	0.107 —
<i>B. The same, with an agricultural interaction alongside</i>			
	Downstream rain (1)	× dam upstream (2)	× agriculture (3)
Irrigated share, 2000	-1.852*** (0.482)	1.501*** (0.564)	-0.402 (0.311)
Cropped-area share, 1990s	-1.738*** (0.456)	1.487*** (0.541)	-0.322 (0.199)
Districts	210, of which 68% have a dam		
District & Year FE	✓		

Notes: Panel A adds an interaction between downstream rainfall (both horizons) and an indicator for any irrigation dam completed by 1999 in the district's upstream catchment, to the four-source specification of Table B.14. Main effects are retained. The moderator is fixed before the estimation window, so it cannot respond to the rainfall shocks being studied, and it is time-invariant, so district effects absorb its level. A dam-release account predicts column 2 to carry the effect — the mechanism needs a reservoir whose release rule can relax when its command area gets its own rain. It runs the other way: the downstream effect is largest where there is no dam and is attenuated where there is one. Dam placement is not random, so column 2 cannot be read as the causal effect of a dam; but the sign is the opposite of the one the hypothesis requires. Panel B asks whether the dam interaction is really an agricultural one: irrigation dams are built where there is irrigable land, so an interaction with dam presence could pick up a larger downstream effect in agricultural districts. Both interactions are entered together, with the agricultural moderator standardised so the main term is the effect at its mean. The dam interaction is unchanged and the agricultural interactions are individually insignificant, so the two are not proxies for one another — the moderators correlate 0.22 and 0.10. Their signs differ: the downstream effect is if anything LARGER in more agricultural districts, which the income reading predicts, and smaller where there is a dam, which the release-rule reading does not.

Table B.22: Upstream rainfall partitioned by dam regulation

	Aggregate	Regulated	Free-flowing
log Upstream Precip. (t)	-0.851 (1.062)		
log Regulated Upstream Precip. (t)		-0.415 (0.872)	
log Free-Flowing Upstream Precip. (t)			-1.178 (1.026)
log Upstream Precip. ($t - 1$)	1.776*** (0.549)		
log Regulated Upstream Precip. ($t - 1$)		1.022*** (0.380)	
log Free-Flowing Upstream Precip. ($t - 1$)			2.402*** (0.884)
Observations	2052	2052	2052
Number of Districts	114	114	114
District & Year FE	✓	✓	✓

Notes: Poisson fixed-effects, UCDP event counts, 2000–2018, districts holding both a regulated and a free-flowing upstream component. Each column enters one component of the same upstream monsoon rainfall; local weather controls (current and lagged) included. The components share 80% of their within-district variation, so the contrast is estimated jointly in the text (total water plus composition) rather than by differencing these columns. Section VII shows the partition has no hydrological first stage and so does not identify buffering. Assam holds 2.7% of the identifying variation; excluding it, the free-minus-regulated gap falls to 0.32 against a placebo 95th percentile of 0.35 ($p = 0.11$). Placebo splits of the free-flowing basins on ten placement characteristics (and a propensity index) never reproduce the real gap, and stratified within-district permutation gives $p = 0.0002$ — direction, not magnitude, is the robust claim. Conley spatial-HAC standard errors (150 km cutoff) in parentheses, which is the paper’s headline estimator and the basis of the significance stars; Online Appendix Figure B.7 re-prices the headline coefficient under alternative variance estimators. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

B.4. The Dam Arm

Table B.23: The dam arm's first stage, and the routing counterfactual

	Free-flowing	Regulated	Gap (SE)
<i>A. Flood-stage days per log point of upstream monsoon rain</i>			
Routed days above the 2-year level	−3.26	13.03	−16.29 (3.20)
Districts		271	
<i>B. CaMa counterfactual: the same rain, dams on vs. off</i>			
Conflict on water that arrived		0.0184	
Conflict on water held back		0.0411 (0.0197)	
Districts		162	

Notes: Panel A regresses routed flood-stage days on log upstream monsoon rainfall separately for catchments with no dam (free-flowing) and with at least one (regulated), with district and year fixed effects; the gap is the difference. A buffering account requires the regulated catchment to deliver *less* flood per unit of rain; it delivers more, because dams sit on the largest rivers — the observational split has no usable first stage. Panel B holds the river fixed: CaMa-Flood is run twice on identical rainfall, with and without reservoir operation, and conflict is regressed on the water that arrived and the water held back, jointly. Attenuated delivery shows no differential conflict response.

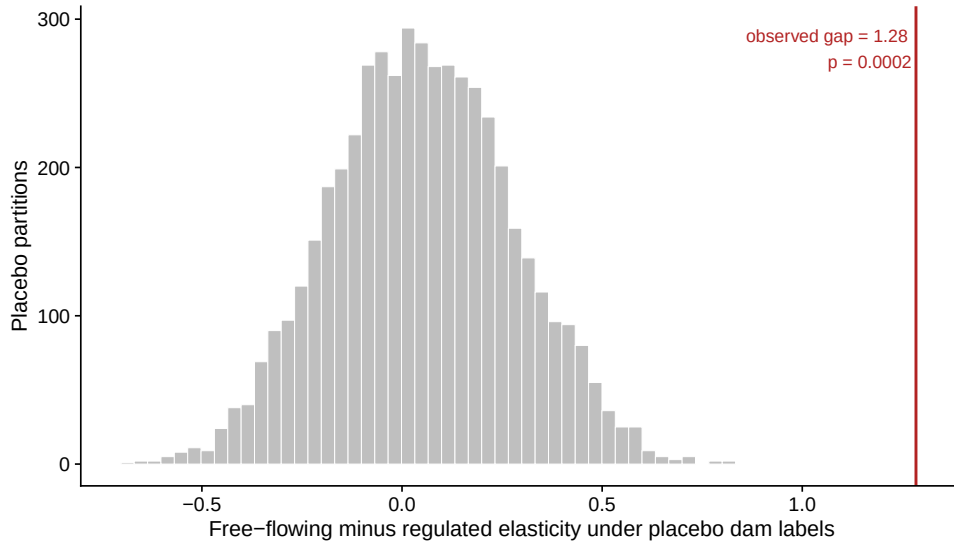


Figure B.13: Permutation distribution for the free-minus-regulated gap, full sample

Notes: Distribution of the gap across 5,000 within-district redistributions of dam labels; the red line is the observed gap. Excluding Assam the observed gap falls to 0.32 and 537 of the same 5,000 placebo draws exceed it.

B.5. Public Employment

Table B.24: The employment guarantee and local rainfall: replication and measure swaps

	Rain, pre-NREGA	× NREGA	Rain, post-NREGA	Districts
His events, his rainfall	−1.402*** (0.276)	+1.298*** (0.329)	−0.103 (0.158)	239
His events, our rainfall	−1.328*** (0.248)	+1.539*** (0.310)	+0.211 (0.161)	239
UCDP events, his rainfall	−0.632 (0.413)	+0.845* (0.493)	+0.213 (0.251)	211
UCDP events, our rainfall	−1.477*** (0.361)	+1.698*** (0.425)	+0.220 (0.234)	211

Notes: [Fetzer \(2020\)](#)’s reduced form — local monsoon rainfall interacted with district NREGA availability, his sample and fixed effects — re-estimated here, then with his TRMM rainfall replaced by this paper’s CHIRPS measure, his news-based event counts replaced by UCDP, and both. The moderation replicates in each: rainfall’s pacifying effect before the programme, absent after. The post-programme interval bounds what can be claimed — it rules out a post effect half the pre-programme size but not a quarter of it, so the programme roughly halves the local rainfall–conflict relationship rather than eliminating it. Standard errors clustered on district.

Table B.25: NREGA and the two rainfall shocks under tightening identification

	Local rainfall		Upstream rainfall	
	× NREGA	Rain, post	× NREGA	Rain, post
<i>A. Post-programme interaction, 1990–2025</i>				
District + year	+0.511(0.394)	+0.178(0.268)	+1.357(0.754)*	+1.189(0.326) ***
+ phase × year	+0.486(0.462)	+0.013(0.283)	+1.538(0.427) ***	+1.125(0.291) ***
+ state × year	+0.436(0.240)*	+0.165(0.324)	+0.728(0.334) **	+0.682(0.364)*
+ state × phase × year	+0.299(0.343)	+0.093(0.315)	+0.601(0.351)*	+0.528(0.454)
<i>B. Staggered roll-out: NREGA × rain within region-year</i>				
Full window	−0.028(0.044)		−0.078(0.062)	
2004–2010	−0.057(0.034)*		−0.082(0.047)*	
2006–2008, staggering live	−0.063(0.033)*		−0.092(0.027) ***	

Notes: Panel A interacts each rainfall source (lagged, logged) with an indicator for NREGA being active in the district, on the fixed-effects Poisson of Table 2, and tightens the fixed effects one step at a time; shown are the interaction and the implied post-programme rain coefficient. Under district and year effects the upstream interaction is large — but “after NREGA” is also simply “later.” Panel B removes “later” entirely: region × year effects with phase-specific rain slopes, so the interaction is identified only from roll-out timing within the same region and year. What survives is near zero for both sources — the programme neither amplifies nor absorbs either shock on this margin. The 2006–2008 window is where phases 1–3 differ within a year; its sample is small and the coefficients are correspondingly fragile. Standard errors clustered on district. The elasticity pair the text quotes (1.344 to 1.252) is the upstream response before and after within region-year, from the same specification as Panel B.

B.6. The Flood Arm

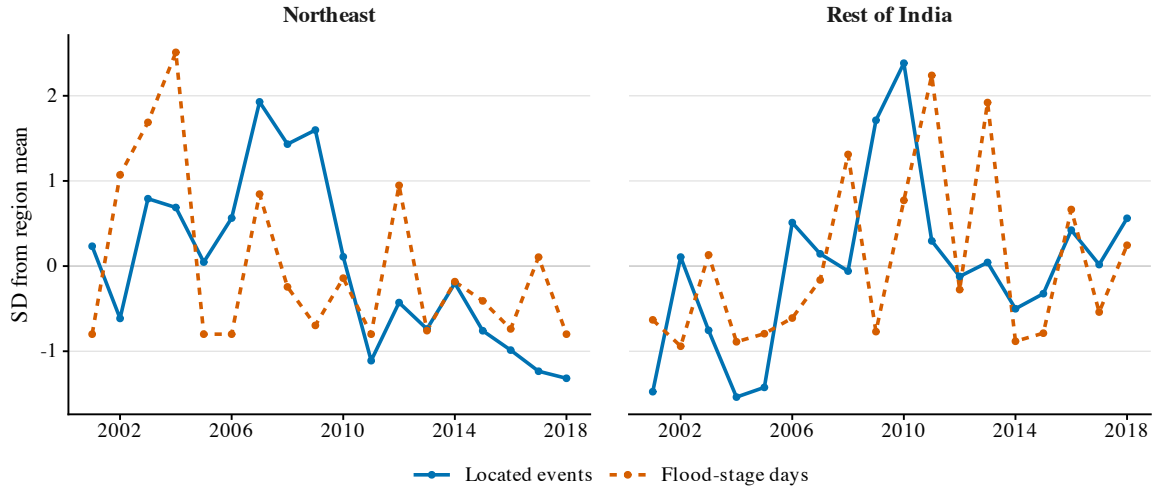


Figure B.14: Regional co-movement of conflict and routed flooding on the estimation sample, standardized within region (one axis; the two series have different units)

Notes: Located events per district (solid) and flood-stage days per district (dashed), 2001–2018. Jammu & Kashmir is omitted: its routed flood series is identically zero. The pooled flood coefficient survives every composition change but collapses under region-by-year effects while nearly all treatment variation survives, and alignment tests (leads, rotation randomization) cannot reject a coincidence of regional cycles; the flood channel’s evidentiary weight rests on the free-flowing versus dam-passing contrast, which shares this regional alignment by construction, and on the main text’s timing evidence.

References

- Adão, R., Kolesár, M., and Morales, E. (2019). Shift-share designs: Theory and inference. *Quarterly Journal of Economics*, 134(4):1949–2010.
- Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1):181–213.
- Emerick, K., de Janvry, A., Sadoulet, E., and Dar, M. H. (2016). Technological innovations, downside risk, and the modernization of agriculture. *American Economic Review*, 106(6):1537–1561.
- Fetzer, T. (2020). Can Workfare Programs Moderate Conflict? Evidence from India. *Journal of the European Economic Association*, 18(6):3337–3375.
- Harari, M. and Ferrara, E. L. (2018). Conflict, Climate, and Cells: A Disaggregated Analysis. *The Review of Economics and Statistics*, 100(4):pp. 594–608.
- Pai, D. S., Sridhar, L., Rajeevan, M., Sreejith, O. P., Satbhai, N. S., and Mukhopadhyay, B. (2014). Development of a new high spatial resolution ($0.25^\circ \times 0.25^\circ$) long period (1901–2010) daily gridded rainfall data set over India and its comparison with existing data sets over the region. *Mausam*, 65(1):1–18.
- Patel, D. (2024). Floods. Working paper, Harvard University.
- Tellman, B., Sullivan, J. A., Kuhn, C., Kettner, A. J., Doyle, C. S., Brakenridge, G. R., Erickson, T. A., and Slayback, D. A. (2021). Satellite imaging reveals increased proportion of population exposed to floods. *Nature*, 596(7870):80–86.
- Yamazaki, D., Kanae, S., Kim, H., and Oki, T. (2011). A physically based description of floodplain inundation dynamics in a global river routing model. *Water Resources Research*, 47(4):W04501.